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Synthetic Travelers: Using LLMs to Generate Realistic Mobility Behaviours

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by

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Master Thesis presented as partial requirement for obtaining the Master's degree in Data Science and Advanced Analytics, with a specialization in Business Analytics

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STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism, any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

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ABSTRACT

Understanding how individuals move through space and time is fundamental to urban planning and quality of life yet collecting the travel data needed to model mobility remains costly and geographically uneven. Classical activity-based models and deep generative approaches often oversimplify individual behaviour and require extensive recalibration when applied to new contexts. This thesis introduces SynTrav, an LLM-agent framework that generates synthetic individual travel diaries from the ODiN single-day self-reported Dutch travel survey. The ODiN population is organized into 190 socio-demographic personas, which condition a Chain-of-Thought pattern extraction stage and a three-step agentic generation pipeline that reasons about each trip given those preceding it. Diaries are evaluated against real ODiN distributions using Jensen-Shannon Divergence across distance, temporal, activity-rhythm, and trip-count dimensions, and benchmarked against a Semi-Markov Process baseline. Transferability is tested internally across held-out Dutch provinces varying in urbanization level, and externally by injecting Portuguese mobility statistics into the prompt to redirect behaviour toward Oeiras without retraining. A human evaluation study with sixty-two respondents, each judging eight real and eight synthetic diaries, finds that evaluators cannot identify synthetic diaries above chance level (46% accuracy, $p = 0.99$), rating them as at least as plausible as real ones. SynTrav outperforms the baseline on temporal coherence and daily trip budgets by larger margins than the baseline leads on distance and activity-purpose distributions. Internal transferability degrades in rural provinces due to underrepresentation in the training data. The external transfer produces directionally correct behavioural shifts across most modes and purposes without retraining. These results position SynTrav as a promising complementary tool for urban mobility analysis in contexts where detailed travel surveys are unavailable or outdated.

KEYWORDS

Travel Diary Generation; Large Language Models; LLM Agents; Human Mobility; Transferability

Sustainable Development Goals (SDG):

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LIST OF ABBREVIATIONS AND ACRONYMS

ABM	Activity-Based Model
AML	Área Metropolitana de Lisboa (Lisbon Metropolitan Area)
API	Application Programming Interface
BFI	Big Five Inventory
BGRI	Base Geográfica de Referenciação de Informação
CoPB	Chain-of-Planned-Behaviour
CoT	Chain-of-Thought
GAN	Generative Adversarial Network
GNSS	Global Navigation Satellite System
GPT	Generative Pre-trained Transformer
INE	Instituto Nacional de Estatística
IPF	Iterative Proportional Fitting
IPU	Iterative Proportional Updating
LLM	Large Language Model
LoRA	Low-Rank Adaptation
LSTM	Long Short-Term Memory
NHTS	National Household Travel Survey
ODIN	Onderweg in Nederland
OSM	OpenStreetMap
PEFT	Parameter-Efficient Fine-Tuning
POI	Point of Interest
RAG	Retrieval-Augmented Generation
SMP	Semi-Markov Process
VAE	Variational Autoencoder

1. INTRODUCTION

Everybody moves through society on a daily basis, being the student catching three different forms of transportation for a two-hour commute to school, the worker stuck in traffic at 8am, or the retiree making their first trip of the day to the grocery store. Understanding how individuals move through space and time is fundamental to urban planning, shaping outcomes that go from traffic congestion, mental health, and energy consumption to overall environmental health and even epidemic control (Y. Liu et al., 2025; Shao, Fan, et al., 2024; Y.-L. Song et al., 2025). Ultimately, accurate mobility modelling can improve quality of life by enhancing all these areas (Y. Liu et al., 2025).

Mobility modelling initially focused on traditional prediction tasks but gradually became central to understanding how populations respond to infrastructure or policy interventions. Evaluating such "what-if" scenarios require travel data under conditions that haven't yet occurred and that no survey can directly capture. Synthetic population generation emerged as way to fill this gap (Agriesti et al., 2021).

Traditional approaches, such as Markov chains and activity-based models (ABMs), have long represented the state of the art in mobility modelling, but they tend to suffer from two main limitations. First, they often oversimplify the complexity of individual behaviour (Amin et al., 2025). While mobility does follow a certain degree of regularity, it is also shaped by complex space-time and social constraints, and is intrinsically tied to personal preferences, social interactions, and environmental factors, dimensions that these models tend to overlook, to their disadvantage (X. Li et al., 2024). Second, these models rely on extensive, high-quality survey data for calibration, which is often expensive and unavailable, and they tend to adapt poorly to change, frequently requiring retraining when applied to a new context. (Amin et al., 2025.; Z. Li et al., 2023; Nadas et al., 2025)

Deep generative approaches such as GANs and VAEs have made considerable progress in mobility generation and can be adequately adapted for this purpose. However, they tend to still overlook the inner mechanisms driving mobility behaviour, lacking semantic awareness of individual intentions, particularly when faced with atypical or nuanced scenarios. (Shao et al., 2024)

LLMs emerge as a promising alternative, showing strong potential to understand these inner mechanisms, summarizing and reasoning about personal mobility patterns and motivations to produce customized behaviour (X. Li et., 2024). This is discussed further in the Literature Review.

This thesis pursues two main goals built on that potential, through SynTrav, the LLM-agent framework proposed in this work. The first is to test whether single-day survey data (ODiN Dataset) is enough for an LLM agent framework to generate synthetic travel diaries that are behaviourally plausible at the individual level. This question is particularly relevant given that the longitudinal, multi-day travel histories used in prior frameworks, while effective, are considerably more expensive and harder to obtain.

The second goal addresses the adaptability of the LLM. Beyond generating realistic diaries for the population they were built on, LLM agent frameworks could, in principle, be redirected toward a different population by adjusting the information given in the prompt, without retraining the model again. This thesis tests that possibility by transferring the framework previously built and calibrated on Dutch data to Oeiras, Portugal, by injecting demographic and mobility statistics from Portuguese sources (IMOB 2017) into the prompt. If effective, this could suggest a path toward lower-cost mobility modelling in contexts where detailed travel surveys are unavailable or outdated.

To this end, this thesis aims to answer the following research questions:

RQ 1: Can a LLM agent framework, conditioned on single-day self-reported survey data, generate synthetic individual travel diaries that reproduce realistic behaviour?

RQ 2: How well does the framework generalize to unseen geographic contexts within the same national survey?

RQ 3: Can prompt-injected demographic and mobility statistics from a different cultural context (Portugal) be sufficient to redirect an LLM agent framework's behavioural prior?

2. LITERATURE REVIEW

This chapter establishes the theoretical foundation of the thesis, first introducing the technical mechanics needed to better understand the methodology, then reviewing the state of the art whose findings and limitations motivate the creation of SynTrav.

2.1 Background Information

Human mobility refers to the movement of people through space and time, seeking to describe, model, and anticipate how people travel in their daily lives. It is commonly captured through trajectories, defined as time-ordered sequences of spatio-temporal points, which form the basic representation from which travel behaviour is modelled. (Kapp et al., 2023)

To model or study human mobility, mobility datasets are required, tendentially coming from two main sources. The first is collected location data, such as GPS traces from mobile devices, which advances in mobile and sensor technologies have made increasingly available. Although, this data raises privacy concerns, and recording the purpose or context of a trip. The second source is travel surveys, in which participants log each trip in a travel diary, typically its start and end time, origin and destination, mode, and purpose, together with their household and socio-demographic details. Travel surveys can be longitudinal or single-day, as in the dataset used in this thesis. The quality of this surveys is often compromised by is cost and low response rate. These limitations have motivated the use of simulation-based techniques to generate synthetic travel survey data. (Bhandari et al., 2024)

Synthetic data is data produced by a model rather than recorded from real-world events. The generating model is designed so that the outputs preserve selected statistical and structural regularities of the source data with the goal of mimicking the real data without copying any individual observation. (Bhandari et al., 2024; Nadas et al., 2025)

Large language models have recently emerged as a generative approach to this problem, and form the methodological core of this thesis. (T. Liu et al., 2024a; Nadas et al., 2025) Their potential and applications will be further explored in the state of the art.

2.1.1 Large Language Models

Typical, Large Language Models are defined as transformer-based language models with hundreds of billions of parameters, trained on vast amounts of text data and designed to understand and generate human language (Alammar & Grootendorst, 2024; Nadas et al., 2025). Their definition, however, remains fluid, since the term tends to evolve with new models release. What might be considered large today can be small tomorrow, as researchers are trying to achieve the same performance with smaller model sizes (Alammar & Grootendorst, 2024).

The journey towards LLMs as we know them today began around 1950 (Shannon, 1951) with the mention of a technique called bag-of-words, a method for representing unstructured text as numerical representations, which was later popularised in the 2000s. Even though these experiments were preliminary, the fundamental idea of “how well can the next letter of a text be predicted when the preceding N letters are known”, remains largely unchanged. (Xiao & Zhu, 2025)

Progressing to later developments, the field advanced with Long Short-Term Memory (LSTM) networks, which could understand sequences of data and retain information over long periods. However, a discover that revolutionized the field came in 2017 when (Vaswani et al., 2023) introduced the Transformer architecture in the paper "Attention Is All You Need". Rather than relying on recurrent connections, the Transformer is built around a mechanism called self-attention, which allows the model to weigh the relevance of every token in the input sequence relative to every other token simultaneously. This eliminates the need for sequential computation, making Transformers faster and more scalable than their predecessors. (Alammar & Grootendorst, 2024; Vaswani et al., 2023).

Transformers become the standard architecture for building large language models. They follow an encoder-decoder architecture, where the encoder generates text representations (embeddings) and the decoder receives those embeddings and produces output text. The design later evolved into the development of decoder-only models, which take word representations and generate text autoregressively, building the foundation for generative AI. This architecture was called a Generative Pre-trained Transformer (GPT) for its generative capabilities (now known as GPT-1 to distinguish it from later versions). (Alammar & Grootendorst, 2024)

Generative LLMs take text as input and attempt to complete it. However, beyond simply completing text, they can be instructed to follow directions through a user query prompt. There are two main categories through which LLMs can be accessed. Closed-source LLMs can only be accessed through an interface or API, and as a result, their underlying code is not shared with the user. In this thesis, closed-source models such as GPT-4o-mini and Claude Haiku were used. On the other hand, the main model used is Llama 3.3-70B, which belongs to the open-source category. A major advantage of open-source models is that they are directly managed by the user. They can be used without depending on an API connection, fine-tuned, and run sensitive data. (Alammar & Grootendorst, 2024; T. Liu et al., 2024a).

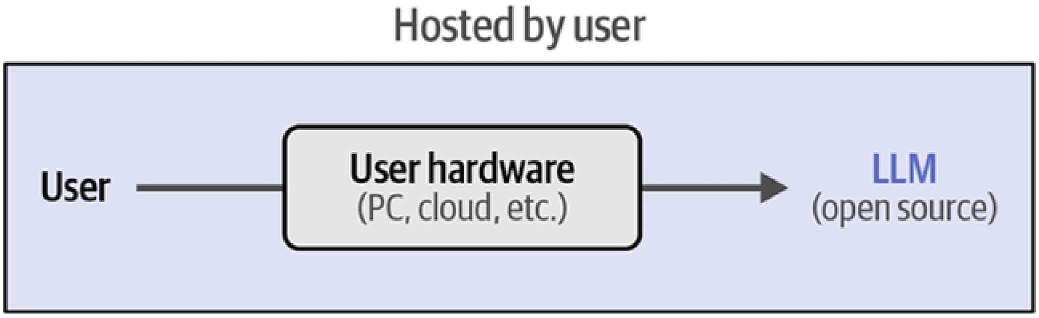


Figure 2.1 – Open source LLMs (Alammar & Grootendorst, 2024)

A fundamental characteristic and advantage of these models is prompt engineering, which refers to the method of providing an LLM with a specific input or cue to generate a desired output or perform a task, directly influencing how effective the models are. Rather than requiring specialists to fine-tune a model for every new task, LLMs exhibit a capability called in-context learning: the ability to learn from examples or instructions provided directly within the prompt, without any parameter updates. This capability manifests in three forms, being them zero-shot, one-shot, and few-shot prompting. Zero-shot prompting, as the name implies, does not use examples, instead, the LLM addresses new problems that were not observed during training. It is possible to adjust prompts to guide the LLM in generating better responses (Alammar & Grootendorst, 2024; Xiao & Zhu, 2025).

There are several strategies to enhance the capability of prompts. One used in this thesis is Chain-of-Thought (CoT) prompting, which encourages the model to reason through a problem before producing an answer, rather than responding directly. The model reasons step by step, where each task is broken into a sequence of intermediate reasoning steps, each conditioning the next (Alammar & Grootendorst, 2024; Xiao & Zhu, 2025).

The ability of LLMs to perceive context and make decisions makes them promising for constructing large language model-powered agents. Many real-world tasks cannot be resolved in a single step, they require the model to plan a course of action, execute it, and adjust based on what has occurred so far, retaining memory of previous steps to inform later ones. This leads to the agent paradigm, in which a language model is no longer treated as a static text generator but as the reasoning core of an autonomous agent capable of perceiving its context, planning, and acting over multiple steps (Gao et al., 2023; Xi et al., 2023).

Drawing on (Xi et al., 2023) an agent needs three key components: brain, perception, and action. The brain can be understood by analogy to our own, since it stores crucial memories and knowledge while handling reasoning. The perception module is responsible for transforming external inputs so that the brain can process them. In the mobility context, such inputs may include travel survey records, spatiotemporal context, or individual profiles, which are ultimately translated into outputs such as full daily trajectories or transport mode choices. Prompt-based and fine-tuned approaches can both operate at the brain stage yet differ in how they shape its reasoning. The action module expands the agent's capacity to interact with its environment, equipping it to execute concrete actions, use external tools, and respond to environmental changes, allowing the agent not only to produce outputs but also to generate feedback.

In the context of this thesis, the action module corresponds to the generation of each travel decision, where every generated trip is conditioned on those that preceded it. SynTrav therefore adopts the LLM-agent paradigm not as a full tool-using system, but as a structured reasoning loop in which a language model perceives a persona, plans a day, and acts it out one trip at a time.

2.2 Literature Review

2.2.1 Modelling Human Mobility Prior LLMs

Human mobility has long attracted computation modelling, with recurring findings showcasing considerable degree of predictability among the complex human travel choices, despite its apparent complexity (Pappalardo & Simini, 2018). Early theoretical grounding came from the social sciences, the Theory of Planned Behaviour (Ajzen, 1991). It posits that human actions, including travel choices, are shaped by attitudes, norms, and perceived control, making it theoretically possible to approximate and predict them.

At first, the proposed approaches treated mobility primarily as a sequence prediction problem, not accounting deeply for the reasoning behind. Markov chains and probabilistic models were used to capture location transition patterns, representing the human movement as location probability matrix over ((Huang et al., 2023). While these models were very interpretable, they were limited to short-horizon predictions and could not account for long-term preferences, trip purpose, or individual heterogeneity.

A shift occurred with large-scale mobility analysis, allowing to established universal mobility assumptions. (Brockmann et al., 2006), tracking the circulation of banknotes across the United States, were among the first to characterise travel distances quantitatively at a population scale. Subsequent work using mobile phone records (Gonzalez & Barabasi, 2008; C. Song et al., 2010) and vehicle traces (Pappalardo & Simini, 2018) cemented that travel distances and radii of gyration (the distance an individual travels from their “center of mass”, meaning the weighted average of all locations they visit), follow heavy-tailed, power-law distributions with exponential cutoffs. This finding implied that most people concentrate their movement in a small number of familiar locations, with rare but significant trips beyond them.

Mobility assumptions, as the previous mentioned one, struggle to capture the heterogeneity in individual travel patterns. Two neighbours may share the same urbanization but dramatically different routines. Confirming this common knowledge, research connecting social network structure to movement showed that social ties contribute substantially to this variability (Hristova et al., 2016). The mobility model SWIM (Small World In Motion) based its approach in the concept of location preference. The user selects the next destination based on past visit frequency and proximity to the home location, allowing the model to reproduce the tendency to return to familiar places and popular ones (Mei & Stefa, 2009). A complementary approach, DITRAS (Diary-based Trajectory Simulator), introduced the possibility of individuals breaking their routines, allowing the framework for probabilistic departures from established patterns. (Pappalardo & Simini, 2018).These models already capture a more personalized representation of mobility, with its limitations.

Within the growth of the Smart City concept, new challenges arrived creating the need to not only approach mobility as traditional prediction problem but to simulate how a population would react to infrastructure or policy interventions. This urged the use of Agent-Based Models (ABMs), which are now better at modelling and forecasting travel behaviour that is sensitive to socio-economic features and individual's characteristics. An ABM requires a dataset granular enough to represent the living population and it's travelling patterns within the modelled area. (Agriesti et al., 2021).

Numerous methods have been proposed to build this representative synthetic version of the real population. The most common one being the Iterative Proportional Fitting (IPF), which assigns sampling weights to individual records such that the weighted distribution of attributes, like age, gender, income, and others, will match the census distributions data. Iterative Proportional Updating (IPU) extends this logic by weighting households to match person and household attributes, and some literature argues it achieves better representativeness as a result (Agriesti et al., 2021; Hörl & Balac, 2021). An example of the latest method being put into use is within the framework of (Moreno & Moeckel, 2018) where they produced a synthetic population of 4.5 million individuals for the greater Munich metropolitan area. For the allocation phase, considering synthesizing a population has two main phases: optimization (fitting) and allocation, and the first one was with IPU, this framework used Monte Carlo Simulation for the second stage, the actual process of replicating actual agents for the synthetic population using a probabilistic selection. Once a synthetic population of individuals is constructed, gravity models are, often, used to quantify origin-destination flows between zones, linking individual characteristics to spatial demand patterns. (Agriesti et al., 2021; Hörl & Balac, 2021; Moreno & Moeckel, 2018).

Building on this algorithms, full ABM frameworks were formed. For instance, with the objective of developing a complete synthetic population and travel demand model for Paris and Île-de-France, (Hörl & Balac, 2021), designed the framework in a way that it can be totally reproducible using only publicly available data as raw input. At the time, most frameworks required detailed firm-level data or commuting matrices that are rarely publicly available, and their methodology was also often insufficiently documented to allow a full replication. With this study they were able to address this gap. However, limitations were still present, namely, the insufficient entries for correlating socio-demographics with commuting patterns.

SySMo's framework (Tozluoğlu et al., 2022), on the other hand, focused in demonstrating that activity-based simulation could be operationalised at national scale. The study represented 10 million agents in their synthetic population, using the assistant of Random Forest classifiers to assign primary travel modes and activity purposes. A proposal that was validated against the Swedish National Travel Survey. (Somanath et al., 2024), extended a similar logic to the travel diary generation, also in Sweden, but at a neighbourhood level, limited to the city of Gothenburg instead of a national scope. One of the limitations of the first was the lack the routing of agents between the origin and destination, aspect that is addressed. The Gothenburg framework consists of a four-stage approach, started synthetic population initialization by creating digital replicas of residents and households, leveraging the Swedish census data. The second stage is the origin-destination assignment where machine learning models, more specifically Random Forests, are used to predict attributes like car ownership and an agent's "primary status" (whether they are a worker or a student), as well as spatial allocation with the gravity model. Next, on the activity sequencing model generates full daily activity chains using stochastic modelling and probabilistic assignment techniques to determine the order and duration of activities (e.g., home → work→ grocery→ home). Finally, the last stage is multi-modal routing, with these trajectories being simulated on the real OpenStreetMap (OSMnx pyhton library) road network. (Agriesti et al., 2021) contributed with a systematic preprocessing pipeline aimed at reducing the data burden of activity-based demand generation, validated through a case study in Tallinn.

A relevant contribution of these approaches to the LLMs-based ones, particular to the Syntrav framework lies in their systematic identification of which sociodemographic features significantly correlate with activity chain generation. Relevant to inform the grouping stage that precedes persona constitution in LLM-based generation, where individuals must be clustered into coherent profiles. This methodology significance will be further explored in future chapters. Figure 2.2, illustrates the correlation findings.

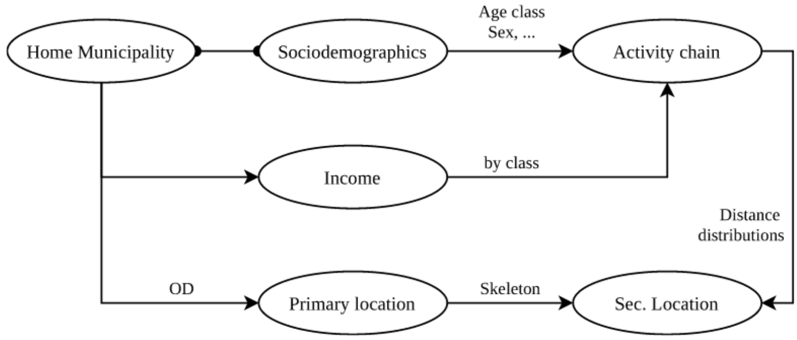


Figure 2.2 – Correlations between the generated attributes and elements of the synthetic travel demand. (Hörl & Balac, 2021)

A critical limitation on this type of approach lies in the dependency on high-resolution data like household surveys, register data or employment records, that is often unavailable due to cost or privacy constraints. Even though, socio-demographic characteristics are being already implemented, the contextual reasoning and some behavioural intent nuances behind travel decisions still remain outside the scope of these models.

Generative deep learning models entered mobility research with the ambition of learning travel patterns directly from data, without the need for prior distributional assumptions. Variational Autoencoders (VAEs) introduced the concept of a continuous latent representation of mobility, enabling the synthesis of diverse trajectories by sampling from that space. Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks demonstrated strong performance in capturing sequential spatiotemporal patterns from GPS trace data (Qin et al., 2025). Generative Adversarial Networks (GANs) took a further step, framing trajectory generation as an adversarial optimisation problem that does not rely on simplified behavioural assumptions and can, in principle, learn richer representations of mobility structure. Despite these demonstrated advances and high prediction accuracy performance, these models are hindered for practical usage due to their limitation in understanding the intentional and contextual aspects of human mobility, especially nuanced and atypical scenarios. (X. Li et all., 2024; Qin et al., 2025)

Looking through these different approaches, it’s possible to observe that there was still a gap centred on understanding the context and motivations behind the mobility choices. Pre-trained Large Language Models represent a different, yet promising resource to address this problem. Having been trained on vast corpora of human-generated text, including travelling and urban information, they demonstrate better semantic interpretability, while still being able to have a constraint aware generation and spatiotemporal constraints (X. Li et all., 2024; Long et al., 2024; Qin et al., 2025).

2.2.2 LLMs AND HUMAN BEHAVIOR

Explaining human choices is a complex task, as performing a behaviour arises from intentions jointly shaped by attitudes and perceived control, which in turn depend on several contextual conditions. Theoretically, the stronger the intention, the more likely a behaviour is to occur. Mobility literature inherited this insight early, treating the intention to travel as a function of individual attributes and situational factors. As discussed in the previous chapter, LLMs are strong candidates for capturing these human nuances (Ajzen, 1991; Nadas et al., 2025).

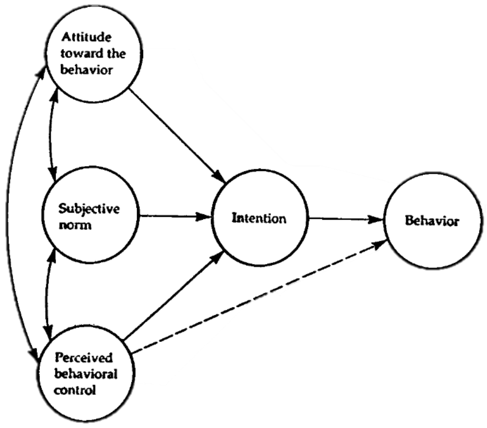


Figure 2.3 – Theory of planned behavior (TPB) (Ajzen, 1991)

Directly inspired by TPB, (Shao, Xu, et al., 2024) introduced the Chain-of-Planned-Behaviour (CoPB) for mobility behavior generation. Rather than prompting an LLM to produce mobility decisions directly, CoPB decomposes the generation task according to the three TPB components, attitude, subjective norms, and perceived behavioral control, while asking the model to reason through each stage before producing an output. This prompting strategy allowed the framework to reduce the error rate of mobility intention generation from 57.8% to 19.4%, representing an improvement over previous baseline on Tencent and ChinaMobile datasets. The authors further explored the synergy between LLMs and mobility rule-based models (e.g. gravity model), demonstrating that token costs could be reduced without sacrificing spatial credibility.

A parallel line of research has investigated whether LLMs can generate coherent agents, armed with consistent profiles, social relationships and behavior tendencies. (Park et al., 2022), through their Social Simulacra framework, demonstrated that LLMs could populate prototype social computing systems with believable synthetic users. Their methodology proceeded in three stages: a persona generation prompt, which produced diverse member profiles with names, backgrounds, and behavioral tendencies; a behavior prompt, that conditioned each agent on their profile and community rules to simulate posts and replies; and a what-if scenario prompt, which introduced hypothetical events and observed how the population of agents responded.

Building on this principle, (Park et al., 2023), deepened the approach with the assumption that agents require conditioning not only on their current environment but also on a vast amount of past experience. To this end, they crafted a persona-based generation incorporating a memory retrieval mechanism in which an agent's past experiences are dynamically updated at each time step and mixed with current contextual inputs when producing new actions. The result was autonomous software agents that can remember past interactions, plan future actions, and react adaptively to changes in their environment. For travel diaries generation, this has direct relevance since a day's decision are inherently dependent on prior activities and time constraints.

A prerequisite for using LLM agents to simulate individual travellers in a reliable way is that the model reflects the specific socio-demographic profiles it is assigned and does so consistently. (Jiang et al., 2024) examined this directly, using GPT-3.5 and GPT-4 to simulate distinct personas defined by the 44-item Big Five Inventory (BFI) personality test. Personas were assigned target scores on each trait and then asked to complete a narrative writing task. The outputs were subsequently evaluated for BFI consistency, demonstrating that both models produced consistent responses across all five personality traits.

The applicability of LLM-based behaviour simulation has been explored beyond only direct social simulation. (Jin et al., 2024) developed a generative driving agent conditioned on post-driving self-reports from human drivers, incorporating expert demonstration data to align generated behaviour with observed human driving patterns. Compared to an unconditioned baseline, the framework with the safety criteria reduced collision rates by 81.04% and increased behavioural similarity to human drivers by 50%, demonstrating an improvement over standard driver simulation approach. In the healthcare domain, Li et al., (2024.) constructed an Agent Hospital by creating a full simulacrum of a hospital environment, in which patients, nurses, and doctors are implemented as autonomous LLM agents interacting across the complete care cycle. Doctor agents improved their diagnostic performance over repeated simulations without requiring manually labelled training data.

Based on the work summarised above, we argue that LLMs are a key ingredient in generating believable agent behaviours, and therefore in modelling more realistic urban mobility. (Jiang et al., 2024; X. Li, et al., 2024; Park et al., 2023)

2.2.3 Mobility Prediction within the LLMs Context

Human mobility prediction studies the forecasting of human movements at individual or collective levels using different datasets and predictive modelling techniques (Qin et al., 2025). In this chapter, we examine how LLMs have been applied in this field. Although human mobility prediction and travel diary generation differ in their genesis, the former has been approached through numerous lenses and has established a set of principles that assist the approach taken in this thesis.

Early work in LLM-based mobility prediction established a paradigm shift away from specific deep neural networks towards general-purpose LLM-based approaches. As a pioneer of this shift, X. Wang et al., (2024) proposed an LLM framework that formats mobility data to capture both long-term habitual patterns and short-term contextual dependencies, while enabling temporally aware predictions. This structured data was embedded into a context-inclusive prompt designed to guide the model to reason sequentially about long-term habits, short-term context, and temporal regularities before producing a ranked list of candidate next locations, each accompanied by a natural-language explanation of the reasoning. The framework was evaluated on two mobility benchmarks: Geolife, which contains GNSS-traced trajectories of individuals, and Foursquare NYC, a venue check-in dataset. The resulting outputs outperformed the accuracy of both classical and deep learning baselines, as well as unstructured LLM prompting strategies, cementing the principle that how an LLM is prompted is as important as which LLM is used. One of the limitations of this study lies in the absence of socio-demographic conditioning and persistent behavioral adaptation

This limitation emerges as one of the primary gaps in maximising the potential of LLMs for analysing human behaviour. LLMs trained on general corpora may carry implicit behavioural biases that misalign with the actual travel choices of real individuals. Both T. Liu et al., (2024) and Mo et al., (2026) sought to address the fundamental question of behaviour alignment, as well as a central concern of this thesis, which is the reliance on extensive GPS tracking data that is often unavailable in practice. T. Liu et al., (2024) addressed these challenges directly in the context of travel mode choice prediction, comparing three prompting strategies: zero-shot prediction, few-shot prediction, and persona-loading, in which the model is provided with a detailed individual profile prior to prediction. Persona-loading outperformed both alternatives, improving the accuracy of mobility prediction and the interpretability of results. This reinforces a methodological principle discussed in previous chapters: conditioning LLMs on individual profiles yields more accurate and behaviourally aligned outputs than unguided prompting. Following a similar line of reasoning, Gong et al., (2024) proposed reasoning through intentions as an intermediate step between individual history and location prediction. Individual mobility histories were encoded as structured textual prompts containing visited points of interest, timestamps, and semantic category labels. The LLM was then prompted to infer latent visiting intentions (e.g. shopping, dining, leisure) before predicting the next mobility decision conditioned on those intentions. A limitation noted by the authors was the difficulty of incorporating age-group and individual-level demographic variation, alongside the persistence of implicit model biases.

The underuse of semantic location names within prompts, token constraints that limit the length of mobility history a model can receive, insufficient trip purpose inference, and the failure to account for the influence of proximity to public transport on destination variability constitute a set of unresolved limitations in mobility prediction. (Qin et al., 2025) introduced the LingoTrip framework to address them. Their central innovation was the use of real-world station and place names as the primary location representation, exploiting LLMs' pre-existing world knowledge. This approach introduces choice variability, since named locations carry semantic associations, for example, a business district implies work-related trips at 9 AM, while a university area implies study-related movement on weekdays. By organising traveller history into long-term habits, mid-term periodic sequences, and short-term recent context, the framework accounts for both routine and incidental behaviours. The approaches described above rely mainly on prompt engineering. (Qin et al., 2026) acknowledged that while prompt engineering improves performance within a given dataset, it limits transferability across different mobility contexts, produces unreliable results in out-of-distribution settings, and is prone to hallucinations when the model is applied outside the geographic or behavioural scope of its training data. In response, they proposed MoBLLM, a foundational model for individual mobility prediction designed to learn a shared representation of mobility behaviour that can be transferred, adapted, and reused across data sources and geographic contexts. Two central shifts were implemented. First, locations are represented using universal integer indices rather than coordinate pairs or place names, allowing the model to learn the underlying dynamics of mobility instead of memorising specific coordinates. Second, training data is structured as instruction-output pairs, enabling the model to be fine-tuned on mobility-specific data in a format that preserves generative flexibility.

The body of work reviewed in this chapter can be summarised around three main principles. First, structured prompt engineering consistently outperforms unstructured counterparts. Second, conditioning the model on individual behavioural profiles (personas) prior to requesting outputs improves both accuracy and behavioural alignment. Finally, reasoning through intentions as an intermediate step enhances the realism of the resulting outputs. These principles are taken into consideration in this thesis, though not applied directly, since the nature of travel diary generation differs from mobility prediction. The goal is not to predict the most likely next location for a specific individual, but rather to generate a plausible sequence of activities, times, locations, and transport modes for a person with a particular socio-demographic characteristics. (X. Li et al., 2024)

2.2.4 TRAVEL DIARY GENERATION

Travel diary generation is a challenging task, fundamentally different from next-destination prediction, since there is no single definitive benchmark against which generated outputs can be compared. The essential goal is to learn realistic individual mobility patterns to generate synthetic travel diary data with characteristics and mobility distributions similar to those observed in real data. A central challenge in this task is preserving true agent heterogeneity, which englobes the diversity of travel patterns that arises from differences in socio-demographic attributes, habits, and contextual circumstances. Since real individuals are too complex to be fully represented, any generative framework must balance the trade-off between model simplicity and the precise modelling of traveller behaviour. (X. Li et al., 2024; Gao et al., 2023)

2.2.4.1 LEVEL FRAMEWORKS OVERVIEW

Travel diary generation has been approached primarily through prompt engineering strategies, fine-tuned LLMs, and, more recently, agentic frameworks. While some frameworks direct their focus to one of these paths, they are not always inherently exclusive of one another. In practice, the boundaries between these approaches are often blurred, and the most prevalent frameworks in the literature are hybrid in design.

Focusing first on prompt-based approaches, several techniques have been employed to bring out the reasoning and planning capabilities of LLMs. A useful distinction can be drawn between zero-shot, one-shot, and few-shot prompting, which rules the number of examples provided to the model and Chain-of-Thought (CoT), which structures how the model thinks by guiding it to process information step by step and build context incrementally before producing the next “thought”. More often than not, these techniques work in pair (Qin et al., 2026).

MobiGeaR (Shao, Fan, et al., 2024) illustrates this combination through introducing a context-aware CoT architecture that incorporates not only travel habits and patterns but also time and date information, based on the assumption that certain activities are temporally implausible, for example most individuals would not go into leisure excursions at 10 PM. Rather than asking the LLM to produce a complete daily schedule in a single step, MobiGeaR decomposes generation into a sequence of individual intention steps, each conditioned on a persona profile that remains constant throughout the process. The persona is constructed by sampling from the attribute distributions of a real-world population dataset, allowing a large population of individually modelled agents to reproduce macroscopic mobility patterns. In the first stage, the model reasons about high-level activity intentions. In the second one, these intentions are mapped to physical locations using a gravity model that allocates destinations based on distance decay and point-of-interest density. This divide-and-coordinate design improved intention accuracy by 62% over deep generative baselines while substantially reducing token cost relative to a fully LLM-driven spatial allocation. CoPB (Shao, Xu, et al., 2024) extends this COT structure further by anchoring the intention reasoning process in the Theory of Planned Behaviour, as discussed in the previous chapter.

(Zhang et al., 2024) introduced MobGLM (Mobility Generative Language Model), a fine-tunning based strategy, which departs from natural language reasoning entirely in favour of direct multimodal sequence modelling. This approach was motivated by the limitations of prior studies in effectively managing spatial data types, including coordinates, zones, geohashes, and points of interest (POIs). The authors reformulate the problem as a multimodal sequence generation task, discretising the day into fixed 15-minute intervals (144 steps per day) where each token encodes four types of information: demographic attributes, activity type, spatial location (administrative zone), and transportation mode. To address the long-tail distribution of non-daily activities (shopping, business trips, and similar), training incorporates a logit-adjustment strategy that amplifies the learning signal for underrepresented activity types, preventing overfitting to the most common patterns.

S. Li et al., (2025) introduced Geo-Llama, an LLM fine-tuning framework designed to enforce multiple explicit visit constraints while maintaining the contextual coherence of generated trajectories. This approach was motivated by persistent limitations in prior work, including training instability, poor scalability with increasing data size, and the absence of control mechanisms to guide trajectory generation under constraints, for example, requiring visits to specific locations. Using Parameter-Efficient Fine-Tuning (PEFT) via LoRA (Low-Rank Adaptation), which modifies a small number of additional parameters rather than the full model, Geo-Llama can directly incorporate visit constraints into the generation prompt, requiring the model to include a specific location within a predetermined time window. This ensures that generated diaries are not only statistically plausible but physically feasible within urban constraints.

Turning to agent-based approaches, which are inherently hybrid in design, the goal shifts from pattern recognition towards treating travellers as autonomous agents with unique personas and motivations. (J. Wang et al., 2024) was among the first frameworks to operationalise this paradigm for mobility, processing semantically rich datasets (e.g., personal check-in data) to produce interpretable simulations of individual daily routines. The study employs two phases: first being the activity pattern identification and second the motivation-driven activity generation. In the first phase, the framework identifies habitual patterns using self-consistency evaluation to ensure that generated behaviours align with historical routines. In the second phase, Retrieval-Augmented Generation (RAG) is employed to retrieve recent and semantically similar historical days as context, partitioning data into historical stays and context stays to capture both long-term habits and short-term dependencies. LLMob outperformed leading generative baselines (including TrajGAIL and DiffTraj) across multiple temporal and spatial metrics, particularly in replicating daily activity routine distributions. It also successfully simulated behavioural adjustments in response to external disruptions, such as pandemic-related policies, using contextual prompts.

Simultaneously, X. Li et al., (2024), demonstrated that ignoring individual profiles leads to the "average person" problem, whereby generated outputs fail to reflect the diversity of real travellers, capturing instead the mean of the population. To address this, they presented MobAgent, a framework capable of generating more realistic individual travel diaries, particularly for individuals who share the same occupation but differ subtly in gender, age, and income level. The framework adopts a two-stage methodology. The first stage, centred on Understanding-Based Mobility Pattern Extraction, grouping individuals through hierarchical clustering based on ten attribute categories (e.g. age, income, and car ownership). After which the LLM extracts group-specific patterns and refines them through self-evaluation using a Chain-of-Thought (CoT) mechanism. In the second stage, each agent engages in recursive motivation reasoning, planning its day step by step as a sequence of intentions before any spatial allocation occurs. To then ensure physical feasibility, the model avoids predicting coordinates directly, instead, MobAgent maps semantic intentions to real-world locations using shortest-path routing on the road network. A notable finding from the ablation studies is that age and income are the dominant drivers of mobility differences, with occupation alone contributing minimally. However, peak performance was achieved when all three attributes were incorporated. Building on these paradigms, (Y.-L. Song et al., 2025) demonstrated their scalability by deploying an LLM-integrated agent simulation across 100,000 synthetic agents in Taipei City, producing full daily schedules and travel chains.

2.2.4.2 Representation of Travel Diaries

A travel diary, as briefly mentioned before, is a set of location-based trajectory point sequences, where each point encompasses a spatial location, a visit time, a travel purpose, a distance, and a transport mode (J. Wang et al., 2024). The encoding choices determine what the LLM can reason about, what input data is required, and what validation against observed data is possible. The following section examines the main perspectives adopted in the literature on this problem.

Beginning with the visit time dimension, MobGLM (Zhang et al., 2024) approached this by transforming traveller activity records into sequences of 15-minute intervals. Each interval was represented by the predominant state capturing the individual's activity, location, and mode of travel during that period. In this case, time is not treated as a variable to be predicted, but rather as a predefined grid that the model fills in. An illustrative example of this grid structure is provided in Figure 2.4.

```
# User's mobility sequences:
["age":55 "Gender":Male "Occupation":Technical_Worker]
"Time stamp:1":["Activity":House "location":103 "Mode":Stay]
"Time stamp:2":["Activity":House "location":103 "Mode":Stay]
...
"Time stamp:16":["Activity":House "location":103 "Mode":Stay]
"Time stamp:17":["Activity":House "location":103 "Mode":Stay]
"Time stamp:18":["Activity":Movement "location":Uncertain_Place "Mode":Railway]
"Time stamp:19":["Activity":Business_Place "location":1121 "Mode":Stay]
"Time stamp:20":["Activity":Business_Place "location":1121 "Mode":Stay]
```

Figure 2.4 – Example of tokenized daily human mobility profile (Zhang et al., 2024)

A less rigid alternative is the day-feature encoding strategy of (Kobayashi et al., 2023), which uses boundary tokens to separate days and feature tokens (e.g., weekday vs. holiday) to capture the periodicity of travel. In total, five distinct features were considered: date, week, holiday, and number of missing entries, collectively characterising the properties of a specific user's day. Analysis of the resulting representations revealed a seven-day cyclic pattern, confirming the expected weekly travel behaviour. Finally, (J. Wang et al., 2024) adopted an approach most closely aligned with that of this thesis, representing time through natural language timestamps, that is human-readable timestamps (e.g., "08:15 AM") and durations expressed as integers in minutes. Treating time as text allows the LLM to reason directly about temporal semantics. For example, it’s able to distinguish between nighttime and daytime, weekday and weekend, or morning peak and evening peak, without requiring these distinctions to be encoded as separate categorical features.

How to represent where an agent goes is one the most fundamental questions when generating a travel diary. MobAgent (X. Li et al., 2024), avoids predicting coordinates directly, instead it maps semantic activity intentions to physical locations through shortest-path routing on a real road network, distinguishing itself from models that depend predominantly on structured GPS trajectory data for both calibration and simulation. This ensures that the generated trajectories respect the geometric constraints of the urban environment but requires access to road network data. LLMob (J. Wang et al., 2024) takes a different approach. Rather than using physical mapping, the framework retrieves candidate locations from a Foursquare POI database based on the user's historical habits and the current temporal context. The LLM then selects the top-k candidate locations from this retrieved list based on inferred motivations, ensuring that every predicted destination corresponds to a real, existing venue in the city. Another popular strategy involves generating intention-time templates, while delegating the final spatial allocation to a gravity model that selects Points of Interest (POIs) based on distance decay and density, an approach adopted by both MobiGeaR and CoPB (Shao, Fan, et al., 2024; Shao, Xu, et al., 2024). Geo-Llama (S. Li et al., 2025) anchors spatial allocation differently, representing trajectories as sequences of visit-level textual tokens with coordinate anchors expressed as Grid IDs.

Activity semantics represents another critical decision, covering how trip purpose is to be encoded. MobAgent (X. Li et al., 2024) treats activity purpose as a natural language reasoning task, employing recursive motivation reasoning in which the LLM agent plans its day step by step, generating intentions as part of a structured narrative. This enables nuanced distinctions, for example between a "school drop-off" and a "work commute. However, the generated purposes are difficult to validate against a rigidly defined dataset. As previously discussed, CoPB (Shao, Xu, et al., 2024) structures purpose generation through the attitude and subjective norm components of the Theory of Planned Behaviour, preventing homogenised schedules by grounding intentions in each agent's social context. . However, this level of abstraction may complicate direct mapping onto transportation classification schemes that prioritise physical destination over underlying motivation. Conversely, MOBGLM (Zhang et al., 2024), foregoes natural language reasoning entirely in favour of multimodal sequence generation, treating activities as mathematical tokens and discretising activity types into fixed categorical classes. While computationally efficient, this approach sacrifices the semantic interpretability of trip purpose. LLMob (J. Wang et al., 2024) uses RAG to retrieve candidate locations from databases such as Foursquare, which provide specific venue category labels(e.g., "Office," "Park," "Cafe"). This grounds activity purpose in real-world categories and ensures alignment between purpose and physical location, however, it relies heavily on the availability of high-quality POI data.

Regardless of the framework, a trade-off is always present. Greater spatial precision comes at the cost of more extensive and resource-intensive data requirements; a richer set of intention categories gains behavioural coherence at the expense of alignment with empirical classification schemes; and when interpretability is prioritised, predictive precision may be compromised.

2.2.4.3 Transferability State of Art

At this point, it is already possible to observe that human mobility modelling has been approached through a wide range of tools and strategies. Nevertheless, a recurrent limitation persists across both mobility prediction and travel diary generation, that is the transferability of trained frameworks to new geographic, demographic, or policy contexts (Qin et al., 2026). This chapter aims to summarise the current state of the art along this dimension.

The transferability problem is not new. Traditional activity-based and agent-based models have long recognized the difficulty of generalizing across geographic contexts. (Agriesti et al., 2021) proposed a methodology for the synthetic population assignment that employs publicly available aggregate data, in order to make it the most replicable and flexible possible. However, as SysMo (Tozluoğlu et al., 2022) mentions, this ABM frameworks have limited generalizability, stating they will have trouble to generalize to other national contexts without full model recalibration. LLMs offer different paths, like prompt contextual injection, instead of adjusting the model weights.

Within the LLM scope, transferability has been directly addressed in the context of mobility prediction. As previously discussed, LingoTrip (Qin et al., 2025) employs semantic anchoring, using actual location names in place of numeric IDs. Because named places carry geographic and semantic associations that the LLM encodes from its pre-training data, the model can reason about locations it has never encountered in training trajectories, effectively handling new users whose past movements are unobserved. Comparing with deep learning baselines, prior models suffered accuracy drops of 7 to 20% when applied to unseen users. Whereas LingoTrip's drop was approximately 1%. However, this transferability was tested to new users within the same region, where behavioural patterns are likely sufficiently similar to those seen during training. MobLLM (Qin et al., 2026), by contrast, demonstrates transferability across entire cities and data sources. Rather than relying on place names, MoBLLM assigns locations universal integer indices, deliberately stripping them of geographic identity. This forces the model to learn the underlying mechanisms of mobility dynamics rather than memorising specific location names. The framework also demonstrated robustness to network disruptions, for example, maintaining predictive performance under metro network changes without retraining. Against state-of-the-art baselines, MoBLLM outperformed the best deep learning model (MHSA) by 50.44% and the commercial LLM-based (LLM-Mob) by 37.30% on standard next-location prediction benchmarks. LLMs emerge as the most promising approach for transferability in mobility prediction, it remains to be examined whether these benefits extend to travel diaries generation.

LLMob (J. Wang et al., 2024) demonstrated how contextual prompt injection can directly shape the behaviour of the generated data. The experiment was due in a pandemic scenario, injecting a prompt that provides the LLM agent with contextual information about specific external circumstances. For instance, a pandemic prompt: “Now it is the pandemic period. The government has asked residents to postpone travel and events and to telecommute as much as possible.” Benchmark that was tested on the same scenario fail to reflect the tendency of each category simulated (e.g. Arts and entertainment, shop & service…). Conversely, LLMob successfully captured the impact of this external factor. In arts and entertainment categories, the generated diaries reflected venue closures and physical distancing guidelines, while in professional activity categories, the framework captured the shift towards remote working.

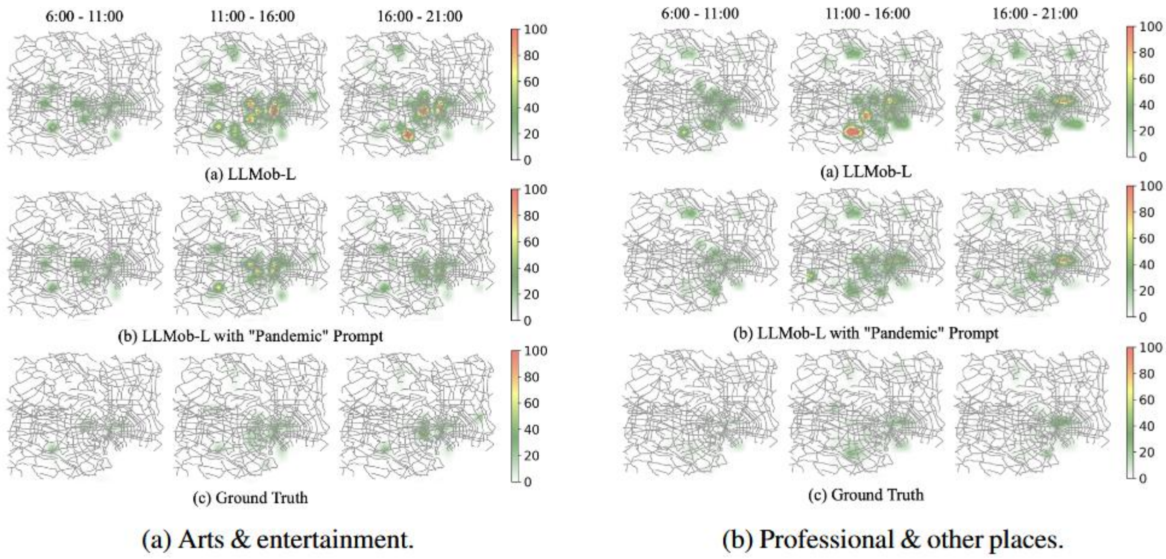


Figure 2.5– Activity heatmaps for the pandemic scenario. (J. Wang et al., 2024)

A comparable experiment was conducted within the HMP-LLM framework (Zhong et al., 2024), employing a two-stage prompting strategy. The input data was first decomposed into three structured components: an instruction element explaining the task, a data element describing the input, and a second instruction element guiding the LLM's reasoning process. These components were combined with contextual information to guide the model's first-stage inference. In the second stage, the model is provided with domain-specific context (in this case, COVID-19 epidemiological statistics) and asked to refine its first-stage output considering this additional evidence. The two-stage design is significant because it separates pattern recognition from contextual reasoning, allowing the latter to be updated independently of the former. (N. Li et al., 2023), investigated the injection of macroeconomic indicators, including annual inflation rate, unemployment rate, and nominal GDP fluctuations, into LLM agent decision-making. Under rule-based and reinforcement learning baselines, the introduction of these macroeconomic inputs produced anomalous outputs, with inflation rate fluctuations exceeding 20% in some scenarios. The LLM-based agents, by contrast, produced stable and numerically plausible economic behaviour without fine-tuning, with inflation consistently remaining within a −5% to 5% range after the third simulation year. While this study is not a direct mobility application, it demonstrates the model's capacity to propagate the behavioural implications of contextual inputs.

Taken together, these findings establishes that LLMs can be adapted to new users, new cities, new policies, and new socio-economic conditions through contextual conditioning, without the model recalibration.

2.2.5 Evaluation Strategies

Having generated the travel diaries, the only remaining step is to compare them against the original data to assess whether the framework was able to capture a sufficient degree of realism. To this end, it is necessary to understand which evaluation metrics have been adopted in the literature for this purpose.

It is unsurprising that metrics commonly used in mobility prediction are inadequate for mobility generation. For instance, Accuracy, Recall, F1 Score, Weighted F1 Score, and the Normalised Discounted Cumulative Gain at rank position k (nDCG@k) (Qin et al., 2025, 2026; X. Wang et al., 2024). The common factor across all these metrics is their reliance on comparison against ground-truth observations. Since synthetic agents do not exist in any real dataset, no ground-truth reference is available against which generated outputs can be evaluated, the adoption of a fundamentally different class of evaluation metrics becomes a necessity.

The most widely adopted evaluation approach across the generation literature is statistical distributional comparison, with the Jensen-Shannon Divergence (JSD) being the most prevalent metric. JSD quantifies the discrepancy between the distribution of characteristics in the generated diary population D′ and the corresponding distribution in the real trajectory dataset D. The measure is bounded between 0 and 1, where a JSD of 0 indicates that the two distributions are identical and higher values indicate increasing divergence. The goal is to minimise JSD across the dimensions being evaluated (Shao et al., 2024a.; S. Li et al., 2025; X. Li et al., 2024.; Shao et al., 2024b; J. Wang et al., 2024).

$$JSD\,(D, D') = \; h\;D + D'\;2 \;-\; h(D) + h(D')$$

Equation (2.1)

To extract the characteristics of the generated and real-world trajectories is necessary to employ other metrics before quantifying the discrepancy between them using the JSD.

The first family of metrics characterises the activity patterns of individual agents. Step Distance (SD) measures the travel distance between each pair of consecutive decision steps within a trajectory, while Step Interval (SI) captures the temporal dimension by measuring the time elapsed between two successive locations on an individual's trajectory (X. Li et al., 2024; J. Wang et al., 2024). The Daily Activity Routine Distribution (DARD), on the other hand, provides insights into how activities are distributed across both space and time, reflecting semantic information such as habitual behaviour. It is computed by constructing a tuple (t, c) at each decision step, where t represents the time interval at which the activity occurs (e.g., indexed from 0 to 144 in a day) and c identifies the activity category based on the location visited at that step. A histogram is then constructed to represent the distribution of these tuples (J. Wang et al., 2024).

To capture the frequency patterns of mobility, a commonly adopted metric is Daily Location Count (Daily Loc), which calculates the number of distinct locations visited per day for each user (S. Li et al., 2025; X. Li et al., 2024; Shao, Fan, et al., 2024). Additional metrics include G-rank, defined as the visiting frequency of the top 100 locations across all users. Transition probability, which measures the probability distribution of a trajectory transitioning from location l_1 to location l_2 across the full set of discretised locations. Also, Distance, which represents the distribution of cumulative travel distance per user per day (Shao et al., 2024a; S. Li et al., 2025).
Examining spatiotemporal patterns more closely, ST Loc captures the number of distinct origin-destination combinations a person uses and at what ((X. Li et al., 2024), while the Spatial-Temporal Visit Distribution (STVD) provides a detailed perspective by assessing the spatiotemporal distribution of visited locations within each trajectory, including geographical coordinates and timestamps (J. Wang et al., 2024). The Radius of Gyration (G-Radius) is also widely adopted, representing the distribution of the spatial range of each user's daily movement. Shifting from individual-level to aggregate spatial assessment, LocFreq partitions the map uniformly into grids, counts the frequency of trajectories visiting each grid, and calculates the JSD between the resulting frequency vectors. ODSim, on the other hand, models the movement of individuals between grids by computing an origin-destination transfer probability matrix and measuring the discrepancy between matrices using the Mean Squared Error (MSE) (Shao et al., 2024a; S. Li et al., 2025).

There is also a need to evaluate the logical consistency of trip purposes. IntentDist captures the number of intentions recorded per day, differing from Daily Loc in that it does not de-duplicate repeated visits to the same location. Turning to semantic evaluation, two metrics are commonly employed: Intention Accuracy (IntentAcc) and Intention Type Distribution (IntentType), referred to in some studies as itdErr and itdType respectively. Applying these metrics requires a preparatory step of matching intention categories to trajectory records. For IntentAcc, since real data inevitably contains uncertainty and randomness, seven days of an individual's trajectories are aggregated into a single representative day through majority voting on each time slice, provided that a full week of trajectory data is available. For IntentType, the time proportion of each intention category within the trajectories is calculated, weighted by duration, and the JSD between the resulting proportion vectors is then computed (Shao et al., 2024a.; Shao et al., 2024b).

3. METHODOLOGY

The code developed in this thesis can be found at this GitHub repository.

3.1 Problem Definition

After the State of Art analysis, it's possible to understand that the usage of LLM's to generate travel diaries is an emerging research area with multiple contributions being published in the last couple of years. Most studies address the systematic reasoning challenge of where, how, when and why people travel in their daily lives by composing agent-based memory architectures, often divided in 2 main phases, pattern identification and travel diary generation with a reasoning mechanism.

Human mobility generation is, at its core, driven by two main factors. The habitual behaviour which encompasses the individual regular activities, characterised by travel frequency, preferred destinations and transportation. The second is understanding the motivation behind these preferences. Collectively, the socio-demographic characteristics as age, income, household composition, employment status, play a huge role in capturing the motivational context underlying individual travel behaviour (Li et al, 2024).

In the generation phase, studies employ the patterns and motivations extracted in the first stage to produce individual-level travel diaries through recursive reasoning mechanisms (Wang et al., 2024; Shao et al., 2024; Chen et al., 2025). Building upon these approaches, this thesis aims to address two main aspects:

Aspect 1. Framework Adaptation

MobAgent (Li et al, 2024) enables the LLM to learn individual profiles and identify associations between individual characters and their mobility behaviours to generate more realistic individual travel diaries through zero-shot learning. Their data was collected via a smartphone app, capturing GPS-traced origin–destination pairs alongside ten individual profile attributes, with multi-day longitudinal records per person. This allows the LLM to examine selected anonymous trajectories and infer which group they belong to, then filling their travel times and intentions. ODiN's self-reported, single-day trip records, even though cheaper to collect, lacks this richness per-person data, making this type of mechanism impractical.

This thesis, therefore, aims to adapt the MobAgent's two-phase reasoning architecture, implementing personas grouping upfront using socio-demographic attributes from the survey rather than derived from past trajectories. The performance of this adapted framework is subsequently evaluated against real ODiN distributions.

Aspect 2. Cross-context transferability

Whether focusing on mobility prediction or full travel diary generation with motivation analysis, a common limitation across current studies is the lack of cross-city generalization, as the existing frameworks are calibrated for individual cities. (Qin et al., 2026) Human mobility patterns differ due to cultural and environmental factors. However, research indicates that over 90% of human activity patterns across different regions are similar. This provides solid foundation for transferring knowledge from data-rich regions to those with limited or incomplete data. (Liao et al., 2024).

LLMob (Wang et al., 2024) demonstrated that contextual modifications introduced at inference time, in their case, a narrative prompt with mobility disruptions during a pandemic (Covid-19), will produce behaviourally coherent shifts in traveling patterns without the need to retrain. Building on this rationale, this thesis aims to inject structural distributional differences, namely modal split, trip purpose profiles, and distance distributions, into the prompt without retraining. Whether this form of statistical context injection is sufficient to redirect the LLM's behavioural priors toward a new cultural population is the research question this work aims to address at this phase.

Transferability will then be evaluated in two experiments: an internal experiment holding out two Dutch provinces within the ODiN dataset, to test generalisation under reduced data but within the same country. The target provinces will differ in urbanisation level. The external experiment will transfer the framework to Portugal using only statistics from the IMOB 2017 survey. A priori, one anticipated and notable difference is bicycle usage, which is very low in Portugal compared to the Netherlands.

3.2 Proposed Approach

As mentioned before, a travel diary, at its core, consists of a sequence of location-based trajectory points, defined as $x = \{s_1, s_2, \dots, s_n\}$, where each point $s_i = (t_i, l_i, e_i, d_i, b_i)$ captures the visit time t_i, spatial location l_i, travel purpose e_i, travel distance d_i, and transport mode b_i of each trip. (X. Li et al., 2024) This is complemented by a set of individual attributes U = $\{u_1, u_2, \dots, u_i\}$, used to assign each individual to their respective persona.

Building upon this foundation, a trip-based representation is adopted in which each diary consists of a time-ordered sequence of trips. Each trip will then be characterised by its origin location o_i, destination location d_i, departure time t_i, transport mode m_i, and trip purpose p_i, capturing not only where the individual was, but the full sequence of the mobility decisions underlying each displacement within the day.

Y = $\{(o_0, d_0, t_0, m_0, p_0), \dots, (o_n, d_n, t_n, m_n, p_n)\}$

Since each ODiN respondent reports a single day, the unit of generation is consistently a single-day diary. Rather than constructing longitudinal sequences per agent, the synthetic population collectively covers the weekly behavioural cycle, as ODiN records are balanced across all days of the week.

SynTrav, the framework proposed in this thesis, is grounded in the LLM agent paradigm, incorporating action, memory, and planning capabilities (J. Wang et al. 2024; L. Wang et al. 2024), and extended with transferability mechanisms. It's structured with four main stages.

Stage 1, Pattern Extraction**,** is responsible for transforming the raw ODiN records into a structured set of behavioural personas and capture the underlying reasons behind observed the mobility patterns. Records are first cleaned and held-out provinces are set aside. In an attempt to preserve behavioural complexity while remaining general enough for transferability, individuals are grouped by the combination of activity status, age class, and income level, producing 190 hierarchical personas. Each persona stores the aggregate statistics of its members, serving as the conditioning signal for all subsequent generation steps. An LLM then applies a Chain-of-Thought (CoT) prompting strategy in a two-step process to produce a natural-language description of each persona's mobility behaviour.

Stage 2, Travel Diary Generation. Upon the defined personas and respective extracted patterns, this stage implements a three-step agentic generation pipeline for each synthetic person, where the output is the individual travel diaries for a weekday.

Stage 3, Evaluation, compares the distributions of the synthetic and real populations using Jensen-Shannon Divergence (JSD) across four metrics. (Li et al. 2024; Somanath, Thuvander, e Hollberg 2024; J. Wang et al. 2024).

Stage 4, Transferability, analysis how well the framework adapts to unseen contexts, within the same cultural setting and across a substantially different one. This is discussed in detail in the section 4.4.

Each stage is described more in depth in the following methodology chapters. An overview of the framework can be found at Figure 3.2.

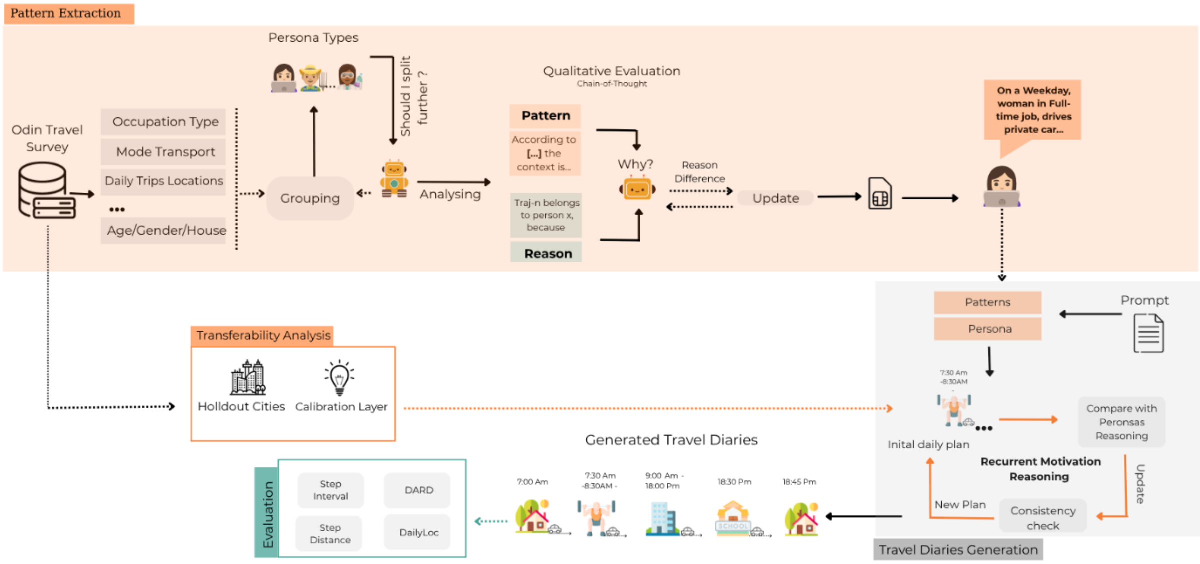


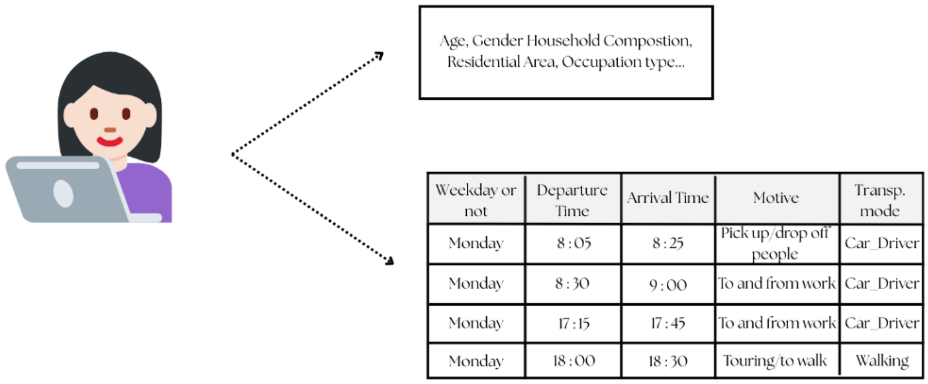
Figure 3.1 - Methodology Overview

3.3 Data Description

Onderweg (En route) in Nederland (ODiN) names the dataset used to generate the travel diary. It's a national travel survey, conducted to provide useful information about the daily mobility of the Dutch population for the benefit of the Ministry of Infrastructure and Water Management since 2018, with information until 2023. However, for this thesis, the focus will be on the available files from 2022, given their further completeness and raw state.

To answer the survey, the participants were asked to report one selected day of their daily lives, detailing where they went that day, for what purpose, by what means of transport and how long it took to get there. Beyond capturing this exhaustive trip-level information, the survey collects socio-demographic characteristics including age, gender, household composition, education level, occupation, and social position. Vehicle-related attributes like car ownership, driver's license possession, and e-bike ownership, along with questions about average transport mode usage, work travel expense reimbursement, and transport poverty indicators are also part of the questionnaire.

Comprising 61,953 respondents across the Netherlands, generating 200,054 individual trip records, with evenly distributed participation across weekdays (ranging from 13.8% to 14.9% per day) and throughout the year (monthly participation ranging from 6.8% to 9.8%) make this 2022 ODiN dataset very valuable to generate a realistic travel diary within the different socio-demographic groups. An example of a respondent trip records from the ODiN dataset it's present at Figure 3.2.



Weekday or not	Departure Time	Arrival Time	Motive	Transp. mode
Monday	8 : 05	8 : 25	Pick up/drop off people	Car_Driver
Monday	8 : 30	9 : 00	To and from work	Car_Driver
Monday	17 : 15	17 : 45	To and from work	Car_Driver
Monday	18 : 00	18 : 30	Touring/to walk	Walking

Figure 3.2 – ODiN Respondent Output Example

3.3.1 Data Cleaning

Some preprocessing steps were required to prepare the data for persona construction and all subsequent stages. The English version of ODiN 2022 was used, in which all variables are stored as numeric or coded values. A metadata codebook provided the column definitions and value mappings. All columns were relabelled with their human-readable descriptions, except those where numeric values were intentional.

Considering that the LLM is instructed to capture what a typical weekday looks like for each persona, atypical days were removed to prevent one-off events from distorting the distributions. Using the ODiN column "Details on reporting day", records flagged as atypical days or no-trip days were excluded. Respondents who reported making no trips but gave a reason (~5.6% of the sample) were retained separately and used downstream to calibrate the probability of a zero-trip day, adding a layer of realism to the generation stage.

The dataset was further filtered to retain only independent trips, meaning each record represents a complete origin-to-destination journey made by a single transport mode, for example, a bus trip to work, rather than a combined bus and walking journey. ODiN supports this directly through the "new move" value in the movement column, which corresponds to exactly one row per trip with its respective mode. This choice simplifies the representation and is consistent with comparable frameworks in the literature (Amin et al., 2025; X. Li et al., 2024; Shao, Fan, et al., 2024)

The dataset was further filtered to retain only independent trips, meaning each record represents a complete origin-to-destination journey made by a single transport mode, for example, a bus trip to work, rather than a combined bus and walking journey. ODiN supports this directly through the "new move" value in the movement column, which

corresponds to exactly one row per trip with its respective mode. This choice simplifies the representation and is consistent with comparable frameworks in the literature

Together, these two filters reduced the original 200,054 trip records to 122,341, removing approximately 38.8% of the data.

3.3.2 Regions of Study

SynTrav is designed to capture group-level mobility patterns defined comprehending who someone is, not where they live. A full-time employed person aged 31–35 with high income travels differently from a retired person aged 65 or older with low income, that difference holds whether they live in Rotterdam or Maastricht.

Consistent with the need to have enough aggregated respondents per demographic group (persona) to produce statistically reliable patterns, the LLM extracts patterns at the province level, the primary administrative unit in ODiN.

Of the twelve Dutch provinces, ten were used for pattern extraction. The remaining two, Utrecht and Friesland, were held out entirely to serve as the internal transferability test. These two provinces were selected deliberately: Utrecht is one of the most urbanised provinces in the Netherlands, while Friesland is predominantly rural, each with enough respondents.

3.4 Personas Creation

Several studies have shown that individual demographic profiles play a critical role in avoiding the "average person" problem in mobility generation. To address It, this thesis constructs a persona object whose purpose is to answer, before any generation begins, the question: who is travelling? To accomplish this, ODiN population is partitioned into groups that are demographically coherent and behaviourally distinct, so the LLM received personalised probability signals.

3.4.1 Data Engineering

Given the high dimensional dataset, some variables were engineered to be more intuitive.

Activity Status represents the occupation of the agent. It was derived from two variables, being them “Paid work OP” and “Unpaid activity OP”, combined with a student card flag. Minors (age < 18) with null activity status were reassigned to student, consistent with Dutch compulsory education law. The final categories are employed, retired, student, homemaker, incapacitated, unemployed, working retired, and inactive.

Age was initially encoded using the original ODiN age bands, but these produced too granular personas that consumed a large number of tokens. The bands were collapsed into nine broader bins: <18, 18–25, 26–30, 31–35, 36–40, 41–45, 46–50, 51–65, and 65+, following the binning strategy adopted by MobAgent. (J. Wang et al., 2024)

Income is not disclaimed with exactitude in ODiN, likely for privacy reasons. Instead, it is encoded across three overlapping variables: decile groups, deviation from the low-income threshold, and deviation from the social minimum. A priority-based function resolved these into six ordered levels, from below subsistence to high income, using standardised household disposable income as the primary source. This measure adjusts for household size by converting income to a single-person equivalent, reflecting actual purchasing power rather than raw household earnings, which would otherwise disadvantage larger households.

Car ownership, originally encoded across nine values, was collapsed into three bins: zero cars, one car, and two or more cars. This consolidation was motivated by the fact that, despite car ownership ranking highest in the LLM scoring, its original granularity produced persona groups that were too small to be statistically reliable, making direct comparison with the other variables unreliable.

3.4.2 Base Groups and LLM Scoring

(X. Li, et al., 2024), demonstrated through an ablation study that diary generation quality improves as attribute granularity increases, from age alone, to age plus income, to the full age–income–occupation combination. Age was identified as the most influential single differentiator, and income second.

Age and occupation formed the base grouping of this Syntrav. Age was selected for its empirically demonstrated importance and because it represents the life stage of the traveller, which is critical in determining daily routines. Occupation, even though it is not the most influential variable in the study previously mentioned, determines the structural organisation of a person's day and serves as the primary split in multiple frameworks.(Agriesti et al., 2021; Somanath et al., 2024; Y.-L. Song et al., 2025; J. Wang et al., 2024)

The next decision was which third variable to add. To assess this empirically, an LLM was prompted with the question: "Does splitting this base group by this variable capture meaningfully different travel behaviours? Assign a score from 1 to 10." This was run across a random sample of 20 base groups × 15 candidate dimensions.

Income ranked 8th overall but was still selected as the third splitting variable. Transport mode frequency variables, including walk and public transportation (PT) frequency, which ranked 2nd and 3rd respectively, were excluded based on the notion that they describe how someone already travels rather than antecedent characteristics that drive travel decisions. Urbanization was retained as a distributional attribute within the persona object rather than a splitting criterion, because anchoring persona definitions to a geographic context would compromise the transferability design.

Before we dive into the final decision, a validation test was conducted to assess whether the LLM scoring reflected genuine subgroup differentiation or was arbitrary. Trip frequency variance, modal split (JSD), and distance distribution (JSD) were computed for the three top ranked, non-selected candidates (car ownership, walk frequency, and PT frequency) and for the selected variable (income).

Car ownership led across all three metrics. Setting it aside, income outperformed walk frequency on all three metrics and outperformed public transport frequency on two of the three (trip frequency variance: 0.0160 vs. 0.0064; distance JSD: 0.0131 vs. 0.0102), losing only on modal split JSD (0.0151 vs. 0.0304). This is consistent with the logic for excluding mode variables, since they only outperformed income on the direct transport-behaviour measure.

Between income and car ownership, the decision required careful consideration. Car ownership received the highest LLM score among the remaining candidates, and the validation confirmed this was not arbitrary. Despite this, income was preferred on two notions: it is more general as a socio-demographic attribute, and car ownership is partially endogenous to income. This is consistent with Amin et al., (2025), who condition vehicle ownership on income, and, aligned with MobAgent's (X. Li et al., 2024) finding that income is the second most important differentiator of travel behaviour, income was chosen as the third splitting variable. Car ownership information will be imputed within the persona object. The full LLM scoring ranking is provided in Appendix Table A1, and the validation chart in Appendix Figure A1.

3.4.3 PERSONA OBJECT

Odin population is split by occupation, age and income, forming 190 personas. A persona does not represent a single synthetic traveler, but a group occupying a shared behavioural possibility space. It provides the LLM with the distributional boundaries of who is travelling, while leaving room for individual variation within generation.

Unlike agent-based models that seed individuals with past recorded personal histories, these probability distributions are the individual-level grounding the LLM receives before any diary is produced. Table 3.1, shows an simplified example of a persona object.

employed | 51-66 | weekday | Income=Above median
1,281 persons, 4,321 trips Mode: 47.7% car, 20.3% on foot, 14.7% bike Purpose: work-trips dominant (37%), then shopping (17.9%) … Timing: bimodal, has a morning spike at 7–9am and an afternoon 5–6pm one Distance: Dispersed, prefers longer trips, 8.6% at 10–15km and 7% at 20–30km Car ownership: 41.8% two cars, only 3.3% no car Urbanization: Spread across urban densities with slight lean toward strongly urban (29.6%) and very strongly urban (23.6%) Demographics: roughly gender-balanced (55.3% male), highly educated (44% university), couples with children (46.4%), couples without (37.5%)

Table 3.1– Persona object output simplified

3.5 PATTERN EXTRACTION VIA CHAIN-OF-THOUGHT (COT)

By extracting the patterns, SynTrav aims to translate each persona's empirical mobility statistics into a natural language behavioral description that the LLM can reason from during diary generation. The prompt used to execute this CoT is present in Prompt A1.

For each of the 190 personas, the pipeline runs two sequential LLM calls:

1. To execute the initial pattern extraction the LLM receives the persona object, which includes the respective aggregated statistics (mode, purpose share…). The model is instructed to write a prose description of the group's mobility behavior. This step establishes what type of traveller the persona represents.

2. The second call refines and validates the behaviors previously extracted behaviors. The LLM receives the initial pattern, eight real individual trajectories, to verify if the pattern describes real people and four masked trajectories, in which one field (mode, purpose, or departure time) is hidden. The LLM is then asked to: flag real trajectories that are inconsistent with the initial pattern, complete the masked fields, and refine the pattern accordingly.

The final output will be the refined pattern on the format of a 3-to-5 sentence natural language description capturing the group's dominant travel habits and characteristics, as shown in the table below.

Employed | 46-50 | Weekday | Income=High income
The high-income, employed individuals aged 46-50 exhibit a refined mobility pattern characterized by a strong reliance on passenger cars for trips to and from work, which account for nearly 30% of their travel purposes, often occurring during peak hours. However, this group also frequently uses non-electric bicycles and electric bikes for commutes and leisure activities, with some individuals engaging in on-foot trips for shopping, sports, and hobbies. Notably, a subgroup within this demographic may have a specific lifestyle involving multiple car trips for picking up/dropping off people, indicating a potential variation in family structure or lifestyle. Overall, their mobility pattern is shaped by their high income, employment status, and household composition, with a mix of sustainable and car-dependent transportation modes

Table 3.2 – CoT pattern example of final output

3.6 Generation Pipeline

The generation pipeline produces a synthetic travel diary for each individual, given their persona profile and the group’s mobility pattern. All generation is zero-shot, since the model sees no examples of completed trajectories and must reason within the conditions described above.

Before initiating the generation pipeline, a zero-trip decision is sampled for each synthetic individual using probability rates computed from ODiN. This aims to preserve the realism that not all agents travel every day. For instance, retired people and homemakers have higher probabilities of staying at home than students or full-time workers.

The pipeline proceeds in three stages:

Stage 1, Motivational Summary. Before any planning occurs, the model is asked to interpret why this person travels. Given the persona's demographic profile and mobility pattern, it produces a short narrative capturing the person's obligations, constraints, and typical weekday travel behavior. This is where the individuality enters. Two people from the same persona group receive the same demographic profile and pattern as input, yet, their motivational summary will differ. This summary serves as reasoning base for the following calls. An ouput example can be found in the Appendix, Table A2.

Stage 2, Daily Pan. The LLM agent analyses the persona’s patterns and the individual’s motivational summary to generate a full day plan from waking up to sleep. The plan is constrained within a trip budget that it’s drawn from a Poisson distribution centred on the group's empirical ODiN mean, and a first departure time anchored to the group's observed morning patterns. This ensures the synthetic day represents the Dutch travel behaviour rather than the model's priors global training priors. The model reasons about which activities to include, whether to chain trips or return home in between, and how to sequence the day consistently with the motivational profile.

Stage 3, Recursive Reasoning Loop. Aided with all the previous gathered information about the traveller, the model is asked at each travel step in the daily plan: “Given everything that has happened so far today, what is this trip about?”. The agent knows at every step where the person has been, what they have done, and what may come next. The travel diary is built trip by trip, with each decision informed by the previous ones, ensuring that purpose and destination have internal coherence across the day.

The final output is a full weekday travel diary in which, for each synthetic person, the LLM produces a sequence of trips with departure time, mode, distance, purpose, destination, and a brief motivation text.

It’s worth noting that SynTrav reasons freely about purpose, motivation and day coherence. Mode assignment, however, is constrained. Each trip's mode is sampled from the empirical ODiN distribution for that persona group, ensuring that mode shares are calibrated to real observed Dutch data. This design choice reflects the absence of individual-level travel histories, from which mode preferences could otherwise be better learned. The same applies for distance, which is filtered by a mode-conditioned physical-plausibility ceiling, for example, cycling capped at 20 km. This mirrors the treatment adopted in comparable frameworks, as in LLMob (J. Wang et al., 2024) where mode and distance are not LLM output fields in the generation schema, with the model focusing instead on semantic reasoning about trip purpose and motivation. The key question is therefore not whether the correct modes or distances appear in the right proportions, but whether the assigned variables make sense given the rest of the trip's attributes and real-world expectations.

Figure 3.3 shows two diary examples, being one from a homemaker and one from a student. The agent successfully inferred the characteristics of each persona and assigned semantically coherent context: the homemaker's diary includes school pick-up and drop-off trips, whereas the student's day is organised around educational activities.

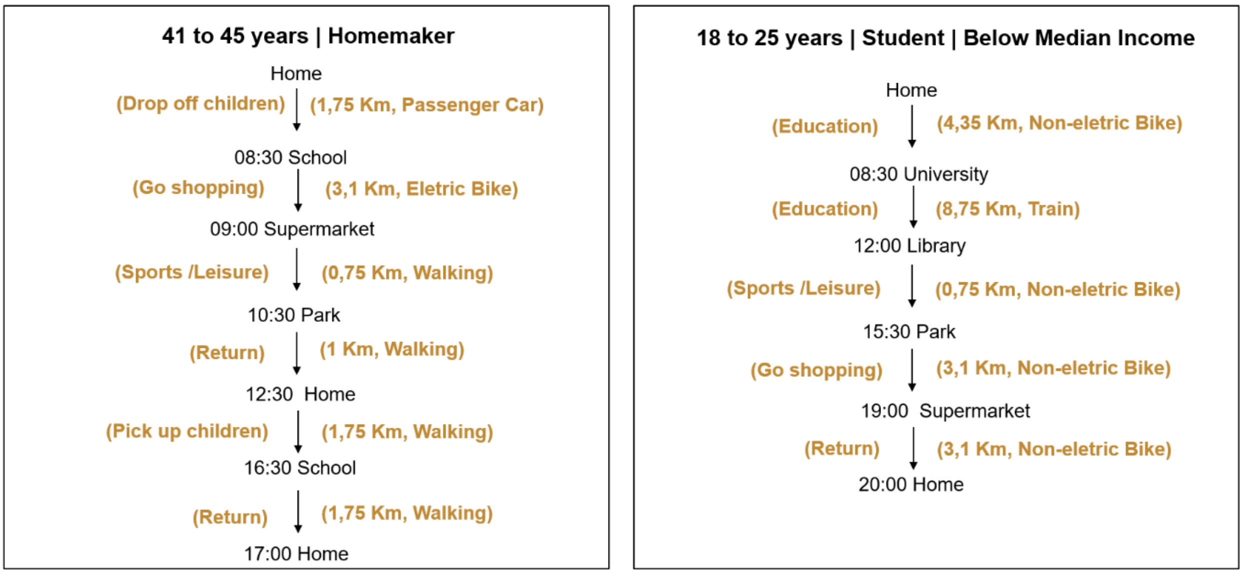


Figure 3.3 – Examples of generated Travel Diary

All prompts used in the generation pipeline can be consulted in Prompt A3, A4, A5. They followed the structure of MobAgent and LLMob, with adaptions necessitated by the different nature of survey data those frameworks were designed for.

3.7 Baseline (SMP)

To understand where the LLM-generated travel diaries offer advantages over a classical statistical approach, a Semi-Markov Process (SMP) is trained on the same ODiN split as SynTrav and evaluated against the same metrics.

Semi-Markov models are well suited to activity-sequence data because they capture both the transition to the next activity and the time spent in each state, allowing duration modelling for inherent different activities such as work and shopping. The SMP is conditioned on persona archetypes, with each component estimated from the observations available within that group and with a fallback to the full-population distribution when the group is too small to fit reliably. (Ma et al., 2009)

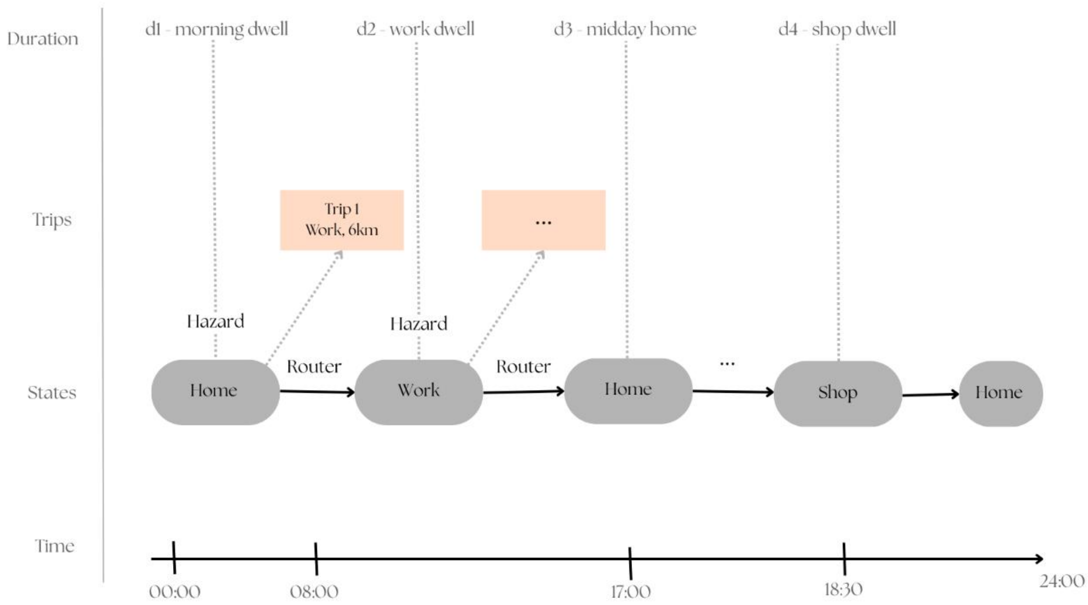


Figure 3.4 – Baseline Overview

The SMP consists of three components. The Router answers "what comes next?" through a transition matrix conditioned on persona and time-of-day bin, which determines the probability of moving to each activity from the current state. The Hazard answers "how long does the agent stay?" by fitting dwell time as a lognormal, Weibull, or gamma distribution per activity type and episode role, with the best fit selected by AIC. Finaly, the Distance component answers, "how far is the trip?" through a log-normal marginal fitted per activity type and persona group. The generation loop runs until the 24-hour budget is exhausted, with a maximum of 20 trips.

3.8 Evaluation

Model performance is assessed by measuring how closely the aggregate distributional properties of the synthetic travel diaries match those of the real ODiN population. All metrics described below are formulated as Jensen-Shannon Divergence (JSD), bounded in [0,1], where lower values indicate closer agreement between synthetic and real distributions. (Shao et al., 2024a.; S. Li et al., 2025; X. Li et al., 2024.; Shao et al., 2024b; J. Wang et al., 2024).

$$JSD\ (P|Q) = \frac{1}{2}\sum_{x} P(x)\log 2\ \frac{P(x)}{M\ (x)} + \frac{1}{2}\sum_{x} Q(x)\log 2\ \frac{Q(x)}{M\ (x)}$$

Equation 3.1

Where P (real population), Q (synthetic population) and M (x) is the average of the two distributions, $M(x) = \frac{1}{2}P(x) + \frac{1}{2}Q(x)$.

3.8.1 STEP DISTANCE (SD)

Captures the distance behaviour ("how far people travel") by measuring the travel distance between two consecutive locations in a trajectory. (X. Li et al., 2024; J. Wang et al., 2024)

$$SD = JSD\ (h_{SD}^{P}, h_{SD}^{Q}\ )$$

Equation 3.2

Where $h^{P}[b]$ is the count of real trips and $h^{Q}[b]$ the count of synthetic trips falling in distance bin k.

3.8.2 STEP INTERVAL (SI)

Measures how much time passes between two consecutive trips made by the same person, capturing whether daily travel is concentrated in tightly spaced trips or distributed across the day. (X. Li et al., 2024; J. Wang et al., 2024)

$$SI = JSD\ (h_{SI}^{P}, h_{SI}^{Q}\ )$$

Equation 3.3

Where $h^{P}[b]$ and $h^{Q}[b]$ are the counts of inter-trip gaps $g_{p,k} = t_{p,k} - t_{p,k-1}$, for $k = 2, \dots, n_p$, falling in interval bin b, pooled across all persons.

3.8.3 DAILY ACTIVITY ROUTINE DIVERGENCE (DARD)

Measures the full day activity rhythm, by capturing the joint distribution of trip purpose and departure time, reflecting which activities people perform at each time of day.

$$DARD = JSD\ (h^{P}_{DARD}, h^{Q}_{DARD}\ )$$

Equation 3.4

where $h^P[b,\pi]$ and $h^Q[b,\pi]$ are joint counts of trips in time bin b with purpose π, flattened to a vector across all $144\ \times \mid \Pi \mid$ cells. (J. Wang et al., 2024)

3.8.4 DAILY LOCATION COUNT (DAILYLOC)

Measures the number of trips made per person per day, capturing daily travel intensity and, under ODiN's structure where each trip corresponds to a distinct destination, approximating the number of distinct locations visited. (X. Li et al., 2024)

$$DailyLoc = JSD\ (h^{P}_{DL}, h^{Q}_{DL}\ )$$

Equation 3.5

where $h^P[b]$and $h^Q[b]$are the counts of persons making b trips on their survey or synthetic day, for $b = 0,1, \ldots ,15$.

3.8.5 SPATIO-TEMPORAL VISIT DISTRIBUTION (STVD)

This metric extends DARD into the spatial domain by measuring the joint distribution of departure time, latitude, and longitude, capturing where and when trips occur across the day. This is the only metric that requires real coordinates, which are assigned through a post-hoc spatial allocation process described in Section 3.9. STVD therefore measures the spatial plausibility of that coordinate assignment process, not LLM spatial reasoning, and is reported separately from the four main metrics. (J. Wang et al., 2024)

$$STVD = JSD\ (h^{P}_{STVD}, h^{Q}_{STVD}\ )$$

Equation 3.6

where $h^P[b,l,o]$ and $h^Q[b,l,o]$ are joint counts of trips in time bin b, latitude bin l, and longitude bin o, flattened to a vector across all $144 \times 20 \times 20$cells.

3.9 SPATIAL ALLOCATION

SynTrav generates travel diaries without geographic coordinates. To assess the synthetic diaries within a real geographic environment, a post-hoc step assigns GPS coordinates to each trip destination, enabling computation of the STVD metric. This step is independent of the generation pipeline and does not influence diary generation and the distributional evaluation.

This allocation adapts from two frameworks: the principle of separating diary generation from spatial grounding, established by MobAgent (X. Li et al., 2024), and the two-level zone-then-POI logic of ASTRA (Schneider et al., 2025). Rather than road-network routing, destinations are assigned by selecting a population-weighted grid cell within the trip's ODiN distance band and sampling a purpose-matched POI within that cell. This approach favours more densely populated areas, for example, an office in a business district will score higher than an isolated one and is therefore more likely to be selected as a destination. Allocation runs sequentially through each agent's day, with each assigned destination becoming the starting point for the next trip, preserving geographic continuity.

Rotterdam was used as the reference city for the Netherlands experiment, where STVD can be computed and compared against the SMP baseline. For Oeiras, the same procedure runs on a Portuguese OSM cache and national population grid. Since no Portuguese travel survey exists for validation, the Oeiras allocation does not produce a comparable STVD score. Its purpose is to demonstrate that the generated diaries can be placed on a real map in a geographically coherent way.

3.10 Human Evaluation Survey

In order to complement the distributional metrics reported in Section 3.8, which assess whether the synthetic population matches aggregate ODiN statistics, but not whether individual diaries are perceived as realistic, a human evaluation survey was conducted.

This evaluation has its roots in the paradigm used to assess whether machine-generated output can be distinguished from human-produced output, adapted here to travel diaries rather than conversation(Jones & Bergen, 2025). It is important to note that this is not a definitive test of realism, but a complement to the quantitative metrics. Specifically, it tests whether individual diaries are plausible to a human reader, regardless of their familiarity with Dutch travel behaviour. The focus, however, is not on the evaluator's ability to distinguish between real and synthetic diaries, but on assessing the perceived realism of the generated output.

Sixteen single-day travel diaries were presented to each respondent: eight real diaries drawn from ODiN and eight synthetic diaries produced by the SynTrav pipeline. Both pools were randomly sampled within their respective groups (seed 42), with basic eligibility filters applied to exclude atypical zero-trip days. The resulting sixteen diaries were shuffled and assigned anonymised IDs (D001–D016) before display, so respondents could not infer a diary's origin from its position in the form. Each diary was displayed as a table of trips, showing start time, travel mode, trip purpose, and distance.

One alignment step was necessary to ensure the two pools followed the same format. ODiN diaries do not record the return home as a separate trip, the respondent's home location is implicit, so the final recorded trip is the last out-of-home activity. The LLM pipeline, by contrast, always generates an explicit return-home trip. To prevent this structural difference from acting as a distinguishing marker, the final return-home trip was removed from each synthetic diary. No other modifications were made to either pool.

For each diary, respondents provided two primary judgements: a plausibility rating on a five-point scale (1 = implausible, 5 = very plausible), and a binary classification of the diary as real or synthetic. When a diary was rated as implausible (below 3), respondents could additionally select one or more reasons from a fixed checklist and leave an optional free-text comment. All respondents evaluated all sixteen diaries. A full example is provided in Appendix B.

A total of 62 respondents completed the survey, yielding 992 diary-level judgements (62 × 16). Before beginning, respondents gave informed consent and reported their age group and familiarity with ODiN. Participation was voluntary and anonymous.

Two analyses were conducted. The primary outcome was discrimination accuracy, measured as the proportion of diaries correctly classified as real or synthetic, tested against the 50% chance level using a one-sided binomial test. Perceived plausibility of real versus synthetic diaries was compared using a Mann–Whitney U test, appropriate for ordinal rating data, with rank-biserial correlation as the effect size measure. All analyses were conducted in Python using the scipy.stats library.

4. EXPERIMENTAL DESIGN

The experiments described in this chapter evaluate the synthetic travel diaries using a population of 570 synthetic agents, derived from 114 demographic personas (five agents per persona) constructed from the ODiN 2022 training set. The personas were originally constructed to cover both weekdays and weekends, yielding 190 groups. However, since it was decided that generation would be restricted to a regular weekday, the personas representing exclusively weekend behaviour were retired, reducing the set to 114. All generation and evaluation is performed on weekday travel behavior. Finaly, performance is assessed using SD, SI, DARD, and DailyLoc.

Every experiment being the Component Ablation Studies, the Reasoning Variants or the model comparison were run with 5 different seeds [3,7,13,27,42] on the same agent population. Metric values are reported as means with standard deviation intervals across the five runs, providing a measure of result stability. When a figure illustrates a single travel diary example, seed 42 is used consistently.

4.1 Experiment 1: Ablation Studies

4.1.1 Framework Component Ablation

The first experiment conducted isolates the contribution of each architectural component to showcase their necessity by evaluating two ablated variants against the full framework. The first ablation “W/o Daily Plan”, removes, as the name indicates, the daily pan component from the generation pipeline to test whether structured temporal reasoning over the full day improves diary quality. The second ablation, “W/o Personas and Patterns”, reduces the agent to a generic LLM with no population-specific context.

4.1.2 Reasoning Variants

The second experiment shifts focus from the framework components to the internal generation design decisions. As discussed in Section 3.6, mode and distance are sampled from the empirical ODiN distribution for each persona group rather than reasoned freely by the LLM. This experiment tests whether that anchoring is necessary, or whether the LLM could produce equally realistic results by reasoning about these attributes itself.

Three variants are evaluated against the full framework, each replacing empirical sampling with LLM reasoning for one factor. The first, "With Mode Reasoning", allows the LLM to select the transport mode rather than receiving a pre-sampled one. The second, "With Distance Reasoning", does the same for trip distance. The third, "With Mode and Distance Reasoning", combines both. In all three variants, the LLM still selects from the persona's ODiN-derived category set, only the selection mechanism changes.

4.2 Experiment 2: Model Comparison

Three reasoning engines were evaluated to assess how much the choice of LLM drives performance and, in particular, whether an open-source model can be competitive with proprietary alternatives. Llama 3.3-70B, GPT-4o-mini, and Claude Haiku were tested under identical framework conditions, prompts, and metrics.

4.3 Experiment 3: LLM Framework vs SMP Baseline

The LLM Framework is compared against the Semi-Markov Proces (SMP) baseline in order to assess where it adds value beyond a statistical approach trained directly on the same data.

Experiments 1 to 3 together address the first research question, evaluating how relevant each component is and how realistic the generated diaries are overall.

4.4 Experiment 4: Transferability

Transferability is assessed at two levels, addressing the remaining research questions.

Internal transferability evaluates whether SynTrav generalises within the same national context, specifically to the provinces of Utrecht and Friesland, which were excluded from persona construction. This experiment will also have the comparison between the LLM and baseline performances.

External transferability evaluates how well SynTrav generalises to Portugal, specifically to the municipality of Oeiras. The central question is whether prompt-injected Portuguese statistics were sufficient to redirect the behaviour of the generated diaries, using patterns learned from Dutch data. As no comparably detailed travel survey exists for Portugal, direct metric comparison as previously done is not possible. The evaluation focus will be on the divergence in generated behaviour between the Portuguese and Dutch synthetic populations across mode, purpose, and distance distributions.

This evaluation differs from the structure of previous experiments in one important respect: the SMP baseline cannot participate in this analysis. As mentioned in chapter 2.5 of literature review, classical models have no mechanism to incorporate external statistics about Portuguese travel behaviour without retraining it on a Portuguese travel survey of comparable complexity.

The prompted used to inject Portuguese statistics can be found at the Prompt 5, in the Appendix.

An additional experiment was conducted to identify which components of the prompt injection influence the observed behavioural shifts. Each context element was tested individually, with conditions ranging from a near-complete absence of location-specific context to explicit behavioural enforcement. To this sensitivity analysis, the set up is a smaller population than all the previous tests, with 228 agents.

The first variant, "No Context", reduces the location injection to a single sentence stating that the persona resides in Oeiras, testing whether the model's mode-choice behaviour can shift without any Portugal-specific grounding. The second variant, "Base Context", reintroduces the IMOB 2017 modal statistics, AML distance tables, and a general narrative about Oeiras, but excludes municipality-specific commuting statistics and any mention of cycling. It tests how well generic grounding alone performs. The third variant, "Full Context", corresponds to Prompt 5 in the Appendix and is the condition used in the behavioural shift analysis presented earlier. It adds Oeiras-specific commuting structure, along with a soft cycling note: "Dedicated cycling infrastructure is minimal; cycling is not a viable mode for most trips." Finally, the fourth variant, "Strict Context", retains all elements of Full Context but adds an explicit instruction to exclude cycling entirely: "Do not generate any cycling trips."

5. RESULTS AND DISCUSSION

This chapter presents the results of the experiments executed. Sections 5.1 to 5.4 address the effectiveness of the framework, built upon ODiN data, as well as the realism of the resulting synthetic travel diaries. The remaining sections address how well the framework performed in the transferability analysis.

5.1 Ablation Studies

5.1.1 Framework Component Ablation

The ablation results in Table 5.1 show that the components do not contribute uniformly across the four metrics, rather, they are complementary. The full framework, with all components integrated, shows the most balanced performance. This is why removing a certain component can slightly improve an isolated metric while overall realism, which requires a joint evaluation across all dimensions, declines.

Table 5.1 – Framework Ablation Studies

Condition	SD	SI	DARD	DailyLoc
Full	0.0227 ± 0.0048	0.0259 ± 0.0044	0.3830 ± 0.0053	0.0421 ± 0.0059
W/o Daily Plan	0.0166 ± 0.0023	0.1315 ± 0.0120	0.5125 ± 0.0151	0.0304 ± 0.0053
W/o Personas & Patterns	0.0228 ± 0.0050	0.1135 ± 0.0064	0.5895 ± 0.0067	0.0986 ± 0.0034

In both ablations, SI and DARD degrade significantly, even when variability is accounted for. These metrics represent the temporal and activity-rhythm structure of the day. Their collapse when parts of the framework are removed confirms that both components are necessary. SD, on the other hand, remains relatively consistent across conditions, with overlapping confidence intervals.

Impact of Daily Plan:

Removing this component, cause a drastic degradation in SI (0.026 to 0.132), with DARD also being significantly affected (0.026 to 0.132). Without the daily plan the recursive loop still conditions each trip on the past schedule but lacks a pre-committed forward plan. Trips are locally plausible but globally incoherent in their temporal rhythm, becoming either too clustered or too spread out across the day. DARD's degradation follows the same logic: without a planned activity sequence, purpose-appropriate trips are still generated but lose their temporal placement, hurting the joint purpose–time distribution. The role of the daily plan it’s comparable to the role that personal trajectory history plays in frameworks with access to longitudinal records.

Notably, removing the daily plan does not worsen DailyLoc, it slightly improves it. However, this is not direct evidence that the daily plan hurts trip-count realism. The more likely explanation is a pipeline fix applied to the full framework requiring every synthetic day to end at home, which shifted trip counts slightly. The ablated version is unaffected by this fix, so the difference reflects the full framework moving rather than the ablation improving. This interpretation is plausible rather than proven, as it rests on a single pre-fix run (seed = 42), where every metric had degraded without the daily plan. Essentially, the metrics the daily plan actually governs (SI and DARD) a show clear improvement in the full framework.

Impact of population-specific context:

Removing the socio-demographic context affects SI and DARD as negatively as removing the daily plan and additionally degrades DailyLoc (0.0421 to 0.0986). Trip frequency and location diversity are tightly connected to persona conditioning. Without population context, the model has no anchor for how many trips a person of a given demographic profile would typically make, and converges toward its pre-trained behavioural prior, which corresponds to no specific real population. Population-specific context is, therefore, the most fundamental layer of the framework. Without it, agents produce generic behaviour and lose the individual diversity that makes synthetic travellers realistic.

The full framework is preferred not because it leads on every isolated metric, but because it is the only configuration that keeps all four dimensions jointly within realistic ranges.

To contextualise SynTrav's DARD score, LLMob (J. Wang et al., 2024) reports values ranging from 0.12 to 0.69 across its models and baselines. SynTrav's 0.383 sits in the mid-range, above the strongest LLM-agent configurations (0.12 to 0.14) but well below the statistical and generative baselines (0.43 to 0.69). The comparison is indicative rather than direct, since the underlying activity taxonomies differ. The score is best read relative to SynTrav's own ablations, where DARD rises from 0.383 to 0.51 - 0.59 as components are removed, confirming that each component contributes meaningfully to the activity-rhythm distribution.

5.1.2 REASONING VARIANTS

The results of the three tested variants can be found in Table 5.2.

Table 5.2 – Reasoning Variants Studies

Condition	SD	SI	DARD	DailyLoc
Full	0.0227 ± 0.0048	0.0259 ± 0.0044	0.3830 ± 0.0053	0.0421 ± 0.0059
With Distance	0.2048 ± 0.0077	0.0243 ± 0.0028	0.3737 ± 0.0032	0.0434 ± 0.0039
With Mode	0.0029 ± 0.0009	0.0264 ± 0.0049	0.3736 ± 0.0064	0.0427 ± 0.0041
With Distance & Mode	0.1709 ± 0.0050	0.0294 ± 0.0039	0.3720 ± 0.0102	0.0440 ± 0.0094

Allowing the LLM to reason freely about distance significantly degrades SD, with the mean rising from 0.0227 to 0.2048, while temporal and activity metrics remain relatively unchanged, which isolates the damage to the attribute being tested. Figure 5.1, which compares the distance distributions of ODiN and the different generated conditions, illustrates why the Step Distance JSD deteriorated so drastically. Even though the LLM still had access to the assigned modes and could only select from the same ODiN distance categories, the generated distances oscillated considerably, with a peak concentrated in short trips and spurious peaks aligned to category midpoints that do not appear in the real data. The model was unable to reconstruct the empirical distance distribution from category labels alone.

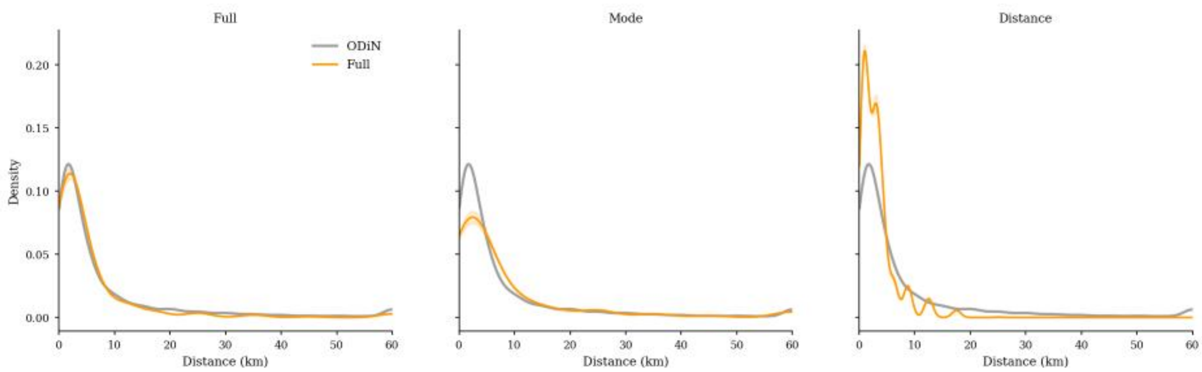


Figure 5.1– Distance Distributions (Reasoning Variants)

At first glance, the mode reasoning variant appears to achieve the most competitive metrics, since it has the lowest SD in the entire study without changing the remain ones. However, examining the mode distributions (Figure 5.2), the attribute the LLM is now reasoning about, reveals a different story. The LLM concentrated trips on car and bicycle, pushing bicycle, which was already high, even higher, while nearly eliminating public transport and suppressing walking. When left to reason, the LLM tends to give more weight to the most common transport modes while neglecting less popular ones.

Notably, the SD improvement is not directly attributable to mode reasoning. The lower value occurs because reducing the walking share (25.4% to 20.0%) reassigns short trips to car and bicycle, which have longer distance distributions, incidentally correcting an over-peaked short-distance baseline in the full framework (53.0% vs. ODiN's 44.8% of trips below 3 km). The long-distance tail is unchanged.

In sum, the four standard distributional metrics are individually insensitive to mode fidelity and should be analysed jointly, since they can move in opposite direction

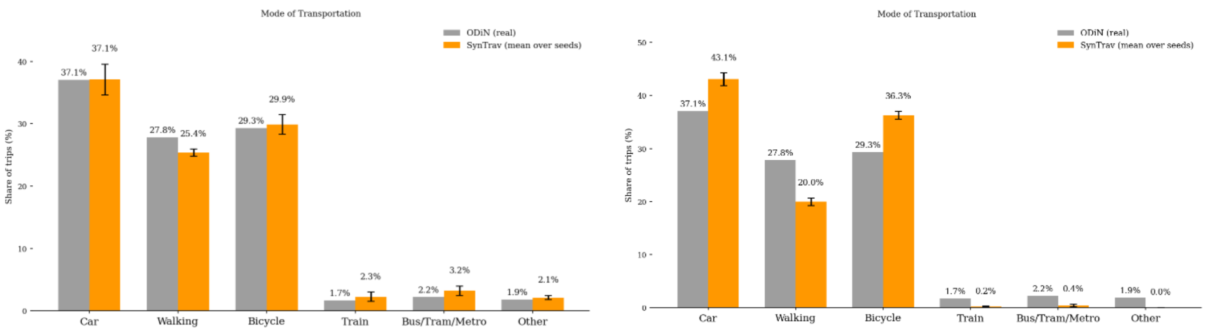


Figure 5.2 - Mode Distribution without Mode Reasoning and with, respectively

The combined experiment behaved as the single-factor results predicted. SD degrades due to the distance-reasoning failure, while the remaining metrics stay stable given their insensitivity to mode fidelity.

Taken together, these variants justify the empirical anchoring adopted in the full pipeline. Distance is anchored because the model cannot reconstruct its distribution from category labels alone, and mode is anchored because reasoning silently trades calibrated mode shares for an illusory gain on a distance metric. In both cases, tying the attribute to observed survey data preserves a fidelity the model does not achieve on its own. The model reasons freely about purpose, motivation, and coherence instead.

5.2 Model Comparison

The performance of the models, under the experimental conditions described in Chapter 4, is summarized in Table 5.3.

Table 5.3– Reasoning Models Comparison

Model	SD	SI	DARD	DailyLoc
Llama 3.3-70B	0.0227 ± 0.0048	0.0259 ± 0.0044	0.3830 ± 0.0053	0.0421 ± 0.0059
GPT-4o-mini	0.0201 ± 0.0047	0.0170 ± 0.0023	0.3700 ± 0.0113	0.0468 ± 0.0067
Claude Haiku	0.0201 ± 0.0019	0.0276 ± 0.0028	0.4016 ± 0.0109	0.0556 ± 0.0058

The three models have competitive performances, with no single model dominating completely and none failing drastically on any dimension. This demonstrates the robustness of the framework itself, since changing the reasoning engine shifts the balance between metrics but does not substantially degrade the overall diary quality.

GPT-4o-mini has the strongest overall profile, with a clear advantage on SI. Its DARD mean is also the lowest, suggesting the best performance on that metric. However, on a closer inspection, the standard-deviation intervals for Llama and GPT-4o-mini overlap on SD (Llama [0.0179, 0.0275], GPT [0.0154, 0.0248]), on DARD (Llama [0.3777, 0.3883], GPT [0.3587, 0.3813]), and on DailyLoc (Llama [0.0362, 0.0480], GPT [0.0401, 0.0535]), indicating comparable performance once variability is accounted for. Claude Haiku is competitive on SD and SI, but falls behind on DARD and DailyLoc, recording the weakest results on the structural metrics. Taking this uncertainty into account, Llama 3.3-70B and GPT-4o-mini are not statistically distinguishable on most dimensions, which in turn reinforces the case for selecting Llama on the grounds of reproducibility and cost

Despite understanding that a proprietary model, specifically GPT-4o-mini, would yield overall better purpose-structure fit, this comes with a trade-off between performance and freedom of usage. As an open source model, Llama 3.3-70B, offers full reproducibility, elimination of API dependency, and lower inference costs (Qin et al., 2026). For SynTrav, whose one of the central contributions is being transferable across regions without retraining, these properties carry more weight than one clear and two modest metric advantages. Llama 3.3-70B was therefore selected as the final model for all generation tasks.

These findings are consistent with previous studies, where GPT-family models (particularly GPT-4o-mini and GPT-3.5-turbo) tend to be the most dominant choice due to the strong temporal coherence while open-source alternatives demonstrate comparable performance when spatial and temporal factors are evaluated together. (Amin et al., 2024.; J. Wang et al., 2024)

5.3 LLM VS BASELINE: DISTRIBUTIONAL FIT AND BEHAVIOURAL REALISM

For a travel diary to be realistic, two conditions must coexist. The first is statistical faithfulness, meaning that we are understanding whether the synthetic population reproduces the distributional properties of real travel data, being departure time peaks, or trip purpose shares. The second is behavioural coherence. The focus shifts to comprehend whether the generated trips make sense within a person's day and not only satisfying statistical targets. Are the sequence of activities coherent having known the persona of the individual, do departure patterns reflect the real world tendencies? Two different frameworks can excel in different areas. This section addresses the first condition, while the following subsection addresses the second.

Looking at Table 5.4, neither the LLM nor the SMP uniformly outperforms the other. The SMP achieves better performance on the metrics more directly tied to trip-level statistics, SD and DARD, while SynTrav outperforms the baseline on SI and DailyLoc, demonstrating better temporal coherence and more accurate daily trip budgets. Notably, the margins by which SynTrav outperforms the SMP are larger than the margins by which the SMP outperforms SynTrav and direction of each win is stable. SynTrav reduces error over the SMP by roughly 63% on SI and 68% on DailyLoc, whereas the SMP's advantage on the metrics it wins is smaller, at roughly 41% on SD and only 13% on DARD.

Table 5.4– LLM vs SMP Baseline

Framework	SD	SI	DARD	DailyLoc
Llama 3.3-70B	0.0227 ± 0.0048	0.0259 ± 0.0044	0.3830 ± 0.0053	0.0421 ± 0.0059
SMP (Baseline)	0.0133 ± 0.0020	0.0694 ± 0.0092	0.3330 ± 0.0059	0.1315 ± 0.0048

Figure 5.3 unpacks these results visually. Trip distances are well approximated by both frameworks, as the scores confirm it. Departure times, however, show a more pronounced divergence. SynTrav is able to recover the characteristic bimodal structure of a weekday, with morning departure peak around 7–8am and an evening return peak around 17–18h, while the SMP does not, keeping the distribution more uniform since it has no prior knowledge of socially synchronized departure times. SynTrav does, however, over-concentrate around these peaks, particularly the morning one, relative

to ODiN. This is likely because the LLM reasons about the canonical version of a weekday rather than the lived one, with all its friction. Aligned with the fact that mid-afternoon and early-evening departures, are more common among retirees, part-time workers, and homemakers, which are underrepresented relatively to workers and students, who dominate the training distribution.

Regarding SI, where the two approaches diverge the most, SynTrav demonstrates superior temporal coherence. Inter-trip gaps in the 30–90-minute range, corresponding to realistic activity durations, are better approximated by SynTrav than by the SMP, which tends to produce longer intervals because it does not reason about the daily structure as whole.

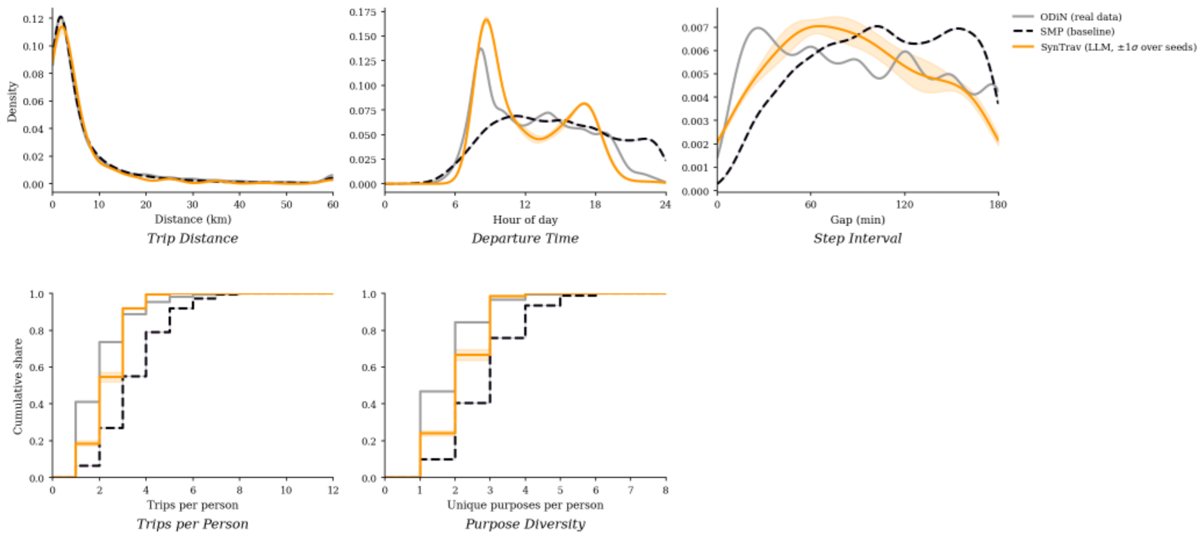


Figure 5.3– LLM vs SMP vs ODIN Distributions Comparison

Looking at Figure 5.4 we can analyse how the trip purpose is being generated in both of the synthetic population through pairing absolute shares with deviations from ODiN. SynTrav errors are concentrated under shopping (+10.1pp), where the LLM over-relies on a generic discretionary label to fill unstructured time, and touring and hiking (−16pp), showcasing a systematic underestimation of Dutch recreational mobility, that happens because the LLM reasons about obligations first (e.g. work, errands, appointments) and recreational trips, carrying no obligation, tend to be omitted from the daily plan. In contrast, LLM presents only an error of +0.4pp for work purpose. The SMP's largest deviations, by contrast, fall on work (−9.7pp), education (−4.8pp) and, also, touring and hiking (-7pp). Work and education are the purposes that more define weekday travel structure.

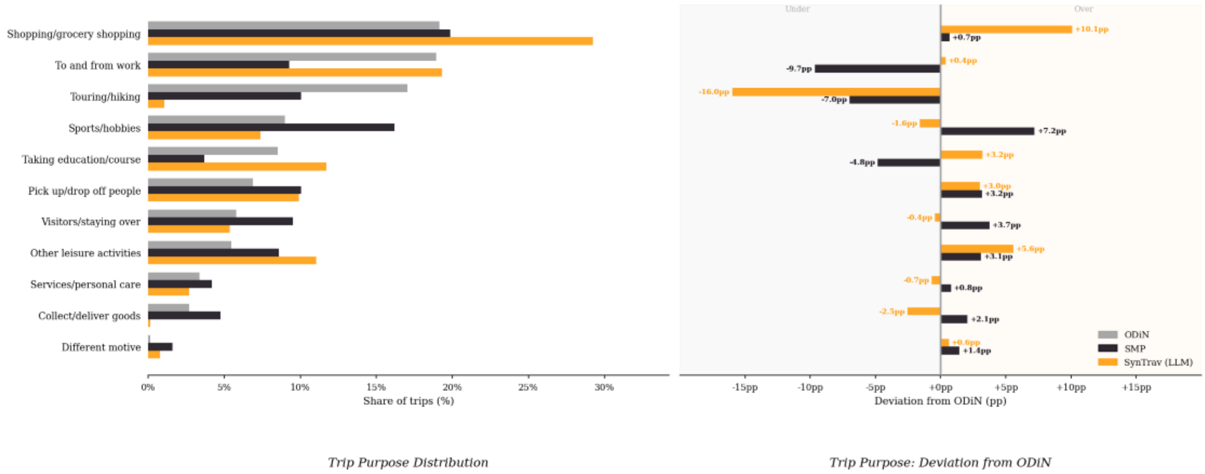


Figure 5.4– Purpose Analysis

Both frameworks have complementary strengths. The SMP performs better on metrics directly tied to fitted distributions, SD and DARD, but at the cost of generating trips with no reasoning about who is travelling or why that end up generating more uninform distributions. SynTrav leads on temporal structure and daily trip budgets and does so by larger margins than those by which the SMP leads. This pattern is consistent with prior work, which found that LLM-generated diaries outperform classical baselines on behavioural realism while classical models tend to retain an edge on numerical fit (Amin et al., 2025). The following section aims to comprehend how coherent and plausible the travel diaries produced by SynTrav actually are.

5.3.1 Behaviour Coherence

Analysing Table 5.5, we can begin to answer whether the produced travel diaries are genuinely plausible and realistic, by assessing whether their behaviour is internally consistent and persona-appropriate. Among non-worker agents, including retired, homemaker, inactive personas, and students, 89.5% (± 1.6) made no work trips, indicating that the LLM correctly suppresses occupational travel for agents without employment status. Among employed agents, 91.8% (±1.6) made at least one work trip, and 99% (±0.8) of students made at least one education trip. These rates do not reach 100%, nor would that be expected since real ODiN records also contain employed respondents who worked from home and students who were absent on their survey day, for example.

In terms of temporal plausibility, it was verified whether the LLM ever scheduled a work trip after an evening leisure trip. This never occurred, consistent with real data behaviour. Another coherence check involved examining distance–purpose consistency, specifically, whether shopping trips cluster at short distances for retired personas, which is the expected behaviour. The LLM captures this correlation but slightly overshoots the distance. Rather than producing a median of 2.00 km (76.1% under 5 km), the LLM produces a median of 3.10 km (56.8% under 5 km). These multivariate patterns are explored further later in this chapter.

Table 5.5– Coherence Metrics (LLM)

Coherence Metric	SynTrav (LLM)
Non-workers, no work trip	89.5 ± 1.6
Workers, ≥1 work trip	91.8 ± 1.6
Students, ≥1 edu trip	99 ± 0.8
Diary ends at home (%)	100.0 ± 0.0
Work trip after 18:00 leisure/visit/sports trip (%)	0.0 ± 0.0
Trips outside plausible distance for occupation×purpose (%)	9.3 ± 0.7

In addition to understand whether the LLM reasons correctly about persona archetype, a temporal analysis by persona was conducted. As shown in Figure 5.5, employed agents and students exhibit a more pronounced bimodal daily pattern compared to retired and homemaker personas, who tend to begin their day in the morning and distribute their trips more evenly, or even reducing them, across the day. SynTrav does not perfectly replicate these patterns, having a tendency to over-concentrate morning departures relative to ODiN, especially regarding students personas. Looking at the working class, the LLM successfully captures the morning routines. The opposite happens with the over-concentration around the evening return time with these two types of personas. Nevertheless, the framework demonstrates effectiveness in capturing distinct daily routines across different socio-demographic segments.

This finding is reinforced by the distance analysis in Figure A2 in the Appendix, which shows that trip length is calibrated to both who is travelling and what they are travelling for. Work trips span the full commute distance spectrum for employed agents and are absent for non-workers. Shopping trips are locally concentrated. Leisure distances are more constrained for retired and homemaker agents, who concentrate most leisure trips within 5 km, while employed agents show a broader spread across all distance bands.

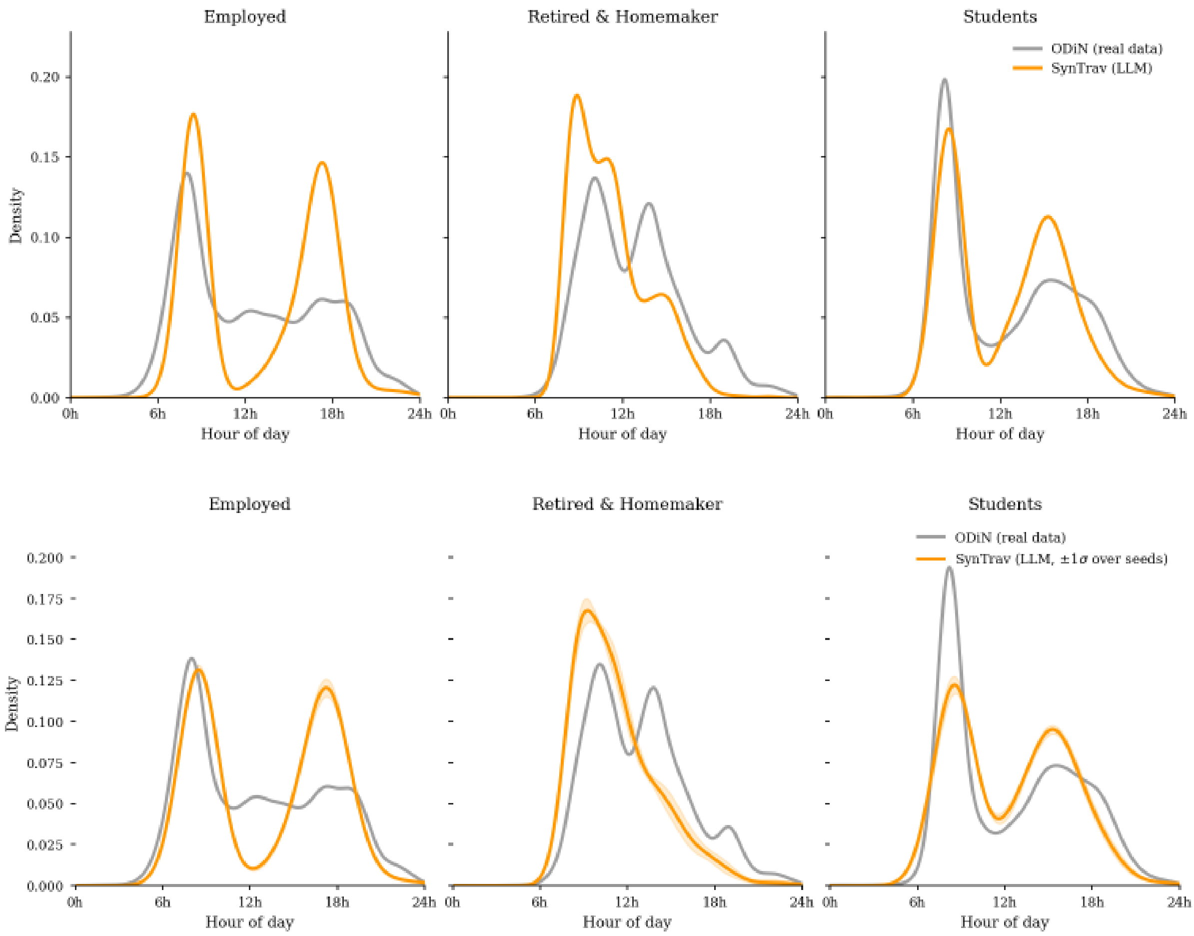


Figure 5.5– Temporal Analysis by Persona

Mode assignment is also plausibly conditioned on the persona type. Figures 5.6 and A3 in the Appendix disaggregate modal split by distance band across occupation groups and income levels respectively. Across all persona types, walking is concentrated in the sub-5 km range, while car use scales progressively with trip length, dominating from 10 km onwards. This pattern holds across income groups. Higher-income persons rely more on the car across both short and long distances, while lower-income persons show lower car reliance, consistent with the lower car ownership rates embedded in their persona objects. Among occupation groups, students show a higher public transport share at longer distances, which reflects real-world behaviour given the high share of students commuting using public transportation. Retires show a preference for short distances and car dominance in longer ones.

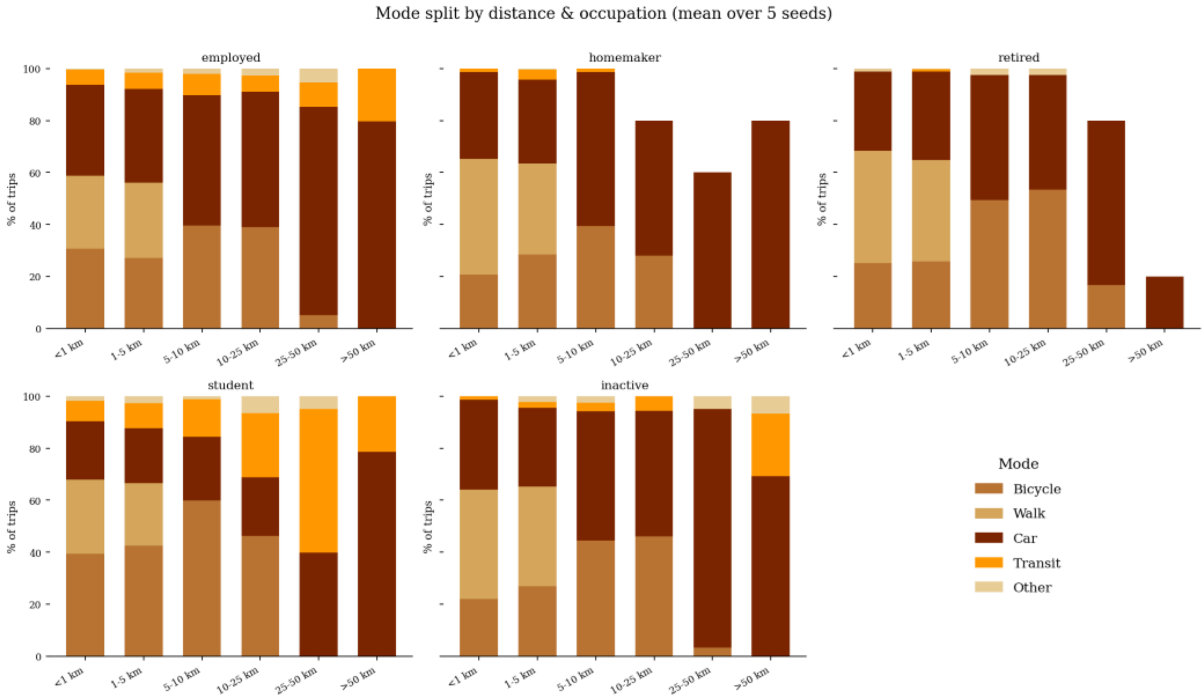


Figure 5.6– Mode split by Distance and Occupation

Taken as a whole, this analysis suggests that SynTrav generates travel diaries that are sensitive to persona attributes across multiple behavioural dimensions simultaneously, remaining plausible and consistent with real-world mobility patterns, even though some limitations persist.

5.4 Human Evaluation Survey

The human evaluation survey gathered sixty-two respondents, each of whom judged 16 travel diaries, rating the plausibility and classifying it as real or synthetic. The survey structure and main design decisions are described in detail in Section 3.10. Here, we observe and analyse the response patterns.

Before presenting the results, it is useful to understand the evaluator pool. Most respondents (≈84%) had little or no familiarity with mobility data or Dutch travel behaviour. This means the survey tests the perception of general readers, specifically, whether an ordinary person can identify irregularities that signal a synthetic diary rather than a real travel routine. The respondents also skewed toward a younger generation, with approximately 45% aged 18–24 and approximately 34% aged 25–34.

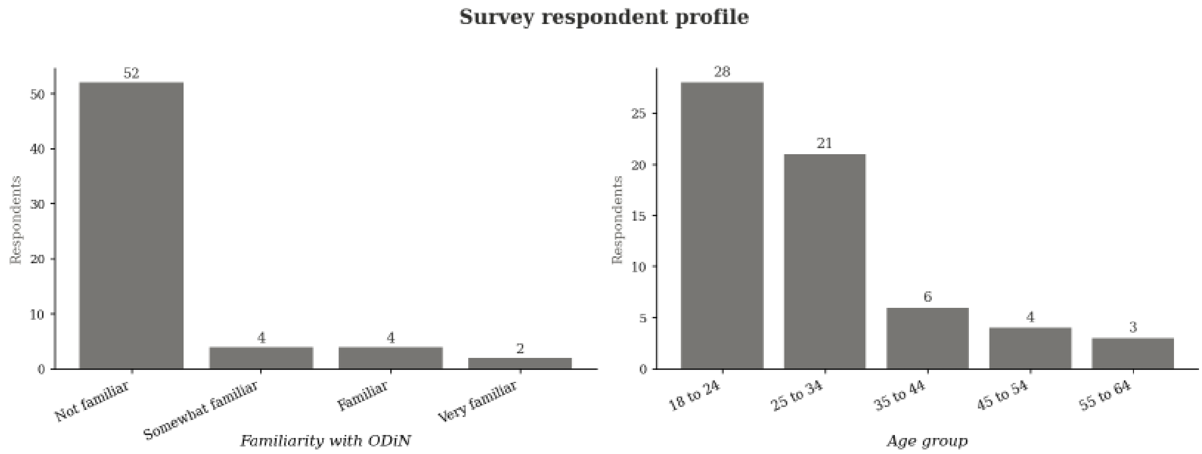


Figure 5.7– Survey respondent profile

One of the main questions this survey aimed to answer was whether respondents could distinguish SynTrav generated diaries from real ODiN ones. Across all 992 judgements, overall classification accuracy was 46% (460/992; 95% CI [43.2%, 49.5%]), below the 50% chance level, as shown in Figure 5.8. A one-sided binomial test found no evidence that accuracy exceeded chance (p = 0.99), indicating that respondents could not identify synthetic diaries more reliably than random guessing. Respondents were in fact more likely to misclassify a genuine diary as synthetic (57%) than to correctly identify a synthetic one (50%). These findings are consistent with Fraser et al., (2025) , that shows that human readers tend to not be reliable detectors of machine-generated content, frequently missing the features that distinguish it.

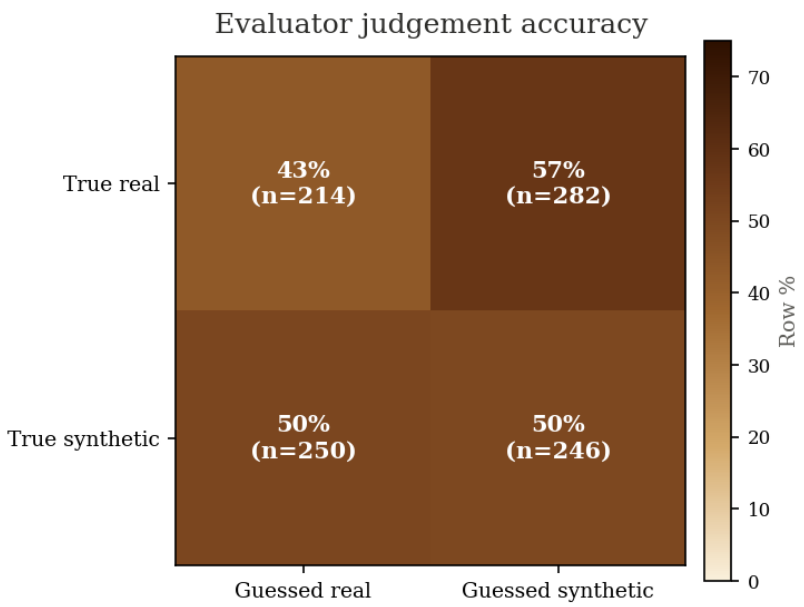


Figure 5.8– Judgement Confusion Matrix

Nonetheless, misclassification was not uniform across diaries. As shown in Figure 5.9, accuracy varied considerably at the diary level. The most convincing synthetic diary (D009) was correctly identified as synthetic by only 19% of respondents, while the most detectable one (D004) was correctly classified by 77%.

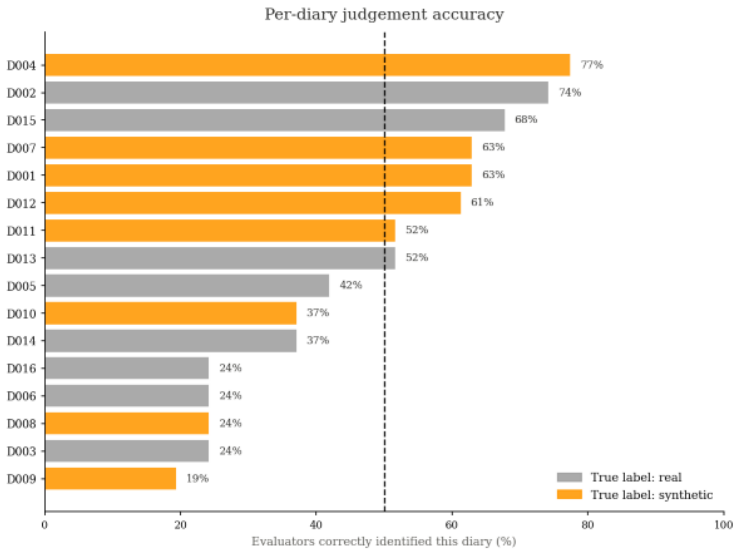


Figure 5.9– Per-diary Judgement Accuracy

Notably, a subset of synthetic diaries was indistinguishable from real behaviour, while a subset of real diaries appeared implausible to evaluators. Figure 5.10 contrasts the most convincing synthetic diary with a real diary that the majority of respondents rejected, illustrating that authentic travel days can be irregular enough to appear synthetic.

Employed · age 65+ · weekday · below median income

Time	Mode	Purpose	Distance
08:00	Car	To and from work	25.0 km
17:30	Car	To and from work	25.0 km

Employed · age 46-50 · weekday · high income

Time	Mode	Purpose	Distance
08:00	Bicycle	To and from work	1.8 km
16:30	Metro	Pick up/drop off people	0.8 km
17:15	Car	Shopping/grocery shopping	87.5 km
18:00	Car	Shopping/grocery shopping	87.5 km

Figure 5.10 -– Travel diaries example (D009 and D006, respectively)

Perceived plausibility was assessed using a rating scale from 1 to 5, where 5 indicates very plausible. Synthetic diaries were rated slightly higher than real ones (mean 3.06 vs. 2.79). A Mann–Whitney test confirmed that this difference is statistically significant but small (p = 0.002, r = 0.11), indicating that synthetic diaries were judged at least as plausible as real ones.

The most frequently cited reasons for rating a diary as implausible were incoherent trip sequences, implausible timing, or an unrealistic number of trips. These reasons were cited with similar frequency for both real and synthetic diaries. However, the reason "purposes don't fit the profile" was cited most frequently for synthetic diaries, suggesting that the primary detectable weakness of the generated output was an occasional mismatch between trip purpose and persona attributes.

These findings can be observed in Figure 5.11.

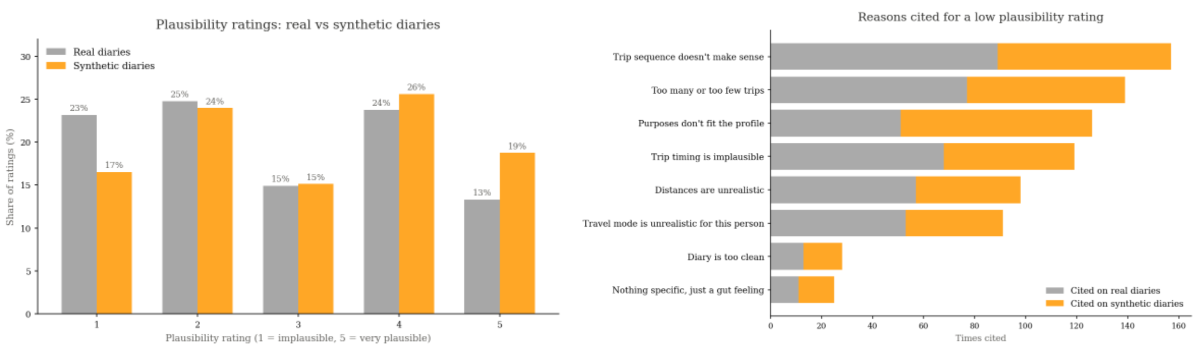


Figure 5.11– Plausibility Analysis

Altogether, evaluators struggled to identify synthetic diaries as generated, rating them at least as plausible as real ones, with persona–purpose consistency emerging as the most reliable cue for synthetic detection.

5.5 Spatial Allocation: Rotterdam

The spatial allocation was applied to Rotterdam to demonstrate that SynTrav can be grounded in a real urban environment, going beyond of understanding behavioural characteristics alone. The allocation method is described in Section 3.9.

This assignment enables computation of STVD, the only metric in the evaluation framework that jointly captures where and when synthetic travellers move across the city. A coordinate-based SD is also computed for additional spatial evaluation. The spatial allocation does not affect the behavioural metrics, its only purpose is to test whether the behavioural structure produced by SynTrav translates coherently into geographic space. To enable meaningful spatial comparison, a larger generation run was performed under identical conditions, yielding 3,210 synthetic individuals per framework. Unlike the five-seed generation used in the main experiments, these runs used three seeds (3, 7, 42). The reduction was motivated by the secondary nature of this evaluation and the larger population size, with time and API cost as practical constraints. Although Llama 3.3-70B is open-source, it is a large model that exceeded local hardware capacity and was therefore run via the Groq API, which carries an associated cost, even if lower than proprietary alternatives.

The results in Table 5.6 show that the SMP achieves a better STVD score, indicating that its assigned destinations are spatially closer to where Rotterdam residents actually travel. This gap was expected, given the SMP's statistical advantage in fitting spatial distributions directly to observed data. To contextualise SynTrav's score, LLMob, which introduced this metric, reports STVD values ranging from 0.38 to 0.58 across its models and backbone LLMs (J. Wang et al., 2024). SynTrav's score of 0.3879 sits at the favourable end of that range, though the comparison is indicative rather than direct, since STVD in LLMob is computed on the model's own generated coordinates rather than a post-hoc allocation. Overall, the score is consistent with a spatial allocation that is functioning as intended, placing synthetic travellers across Rotterdam in spatio-temporally structured rather than random patterns.

Comparatively, both frameworks achieve similar coordinate-based SD. Since both use the same POI allocator and SD is computed from haversine distances between allocated coordinates, this indicates that LLM-estimated distance classes are as effective as the SMP's in guiding the allocator toward geographically plausible destinations.

Table 5.6 – LLM vs SMP Baseline Spatial Metrics

Framework	STVD	SD (coordinates based)
Llama 3.3-70B	0.3879 ± 0.0021	0.0856 ± 0.0010
SMP (Baseline)	0.2732 ± 0.0048	0.0838 ± 0.0012

Figure 5.12, computed on the results of Llama 3.3-70B, demonstrates that the spatial allocation does not produce a uniform distribution of destinations across agents. The destinations adapt to the persona type. Holding age constant at 41–45, employed agents concentrate destinations around office and urban-core locations (Office 38%), whereas unemployed agents of the same age shift toward parks and more dispersed destinations (Park 38%, with office no longer among the leading categories). Across occupation groups more broadly, students aged 18–25 cluster in the city centre with a

dominant share of education destinations (School 54%), while retired agents (51–66) concentrate on parks and local amenities (Park 47%, Market 27%).

A finer illustration of this sensitivity is presented in Appendix, Figure A5. Two employed women aged 31–35, differing only in income level, show distinct destination distributions across Rotterdam on a weekday. Both concentrate around the central and eastern districts, but their activity mix differs with income. The hight income profile allocates a larger share office destination (39% vs 34%) and introduces leisure activities absent from the low-income profile, specifically sports facilities (11%), alongside a stronger northern density cluster. The low-income profile tends to gravitate more towards everyday-service destinations such as markets and schools (Market 27%, School 20%).

These results demonstrate that persona conditioning is sufficient to produce socioeconomically differentiated spatial behaviour. No explicit geographic rule per profile was imposed beyond the urbanisation level in the persona object. The behavioural differences encoded in the persona (e.g. trip purposes, distance classes, and activity sequences) propagate through the spatial allocator and produce distinct destination distributions across the city. The images were produced using the synthetic diary with seed equal to 42.

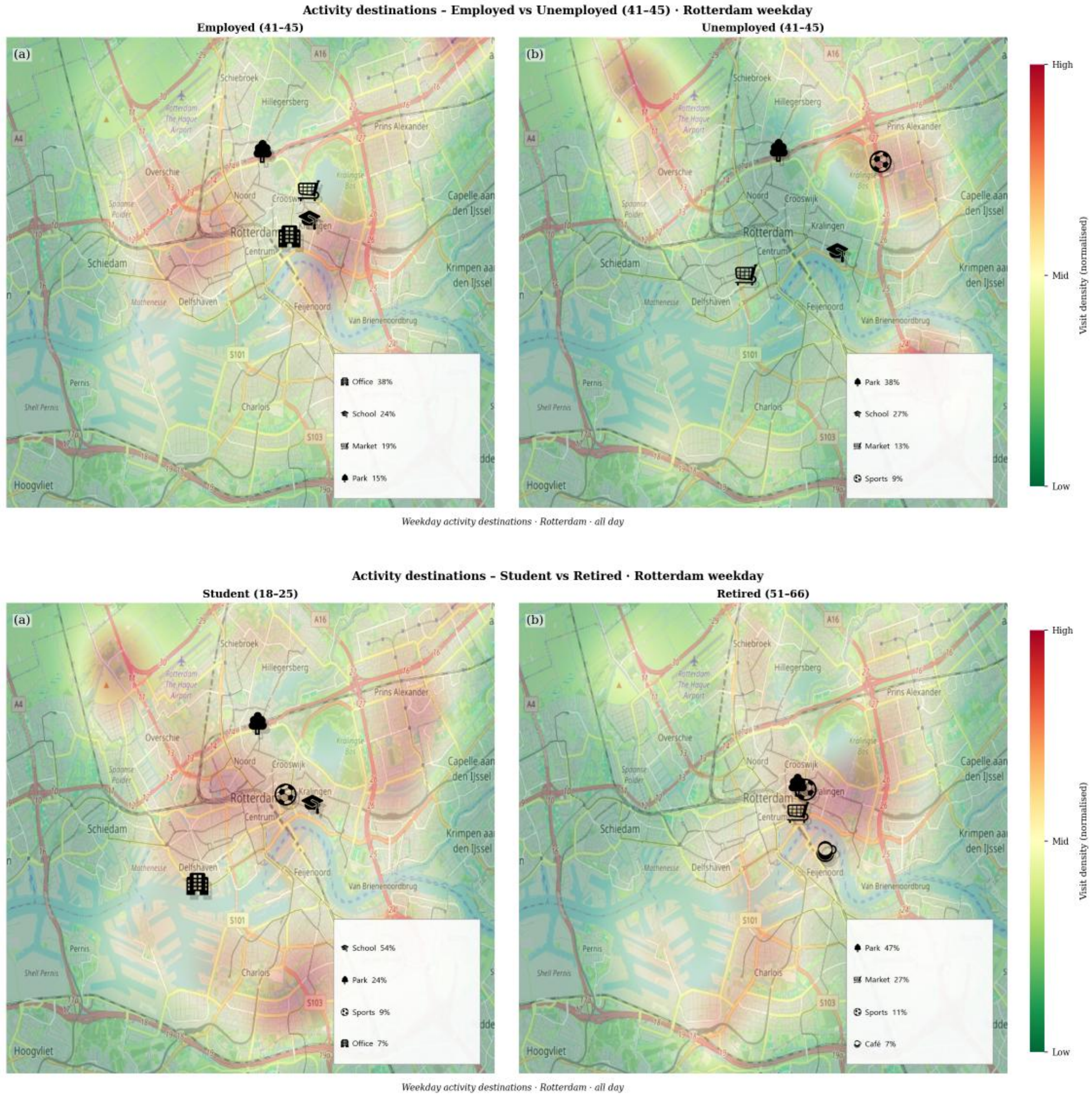


Figure 5.12– Activities Destinations by Socio-Demographic Profiles

5.5 Internal Transferability

Based on Table 5.4, similar patterns to the main evaluation can be observed, with both frameworks performing comparably to or slightly better than the full-province results across most metrics.

Table 5.7– LLM vs SMP Baseline for Utrecht

Framework	SD	SI	DARD	DailyLoc
Llama 3.3-70B	0.0218 ± 0.0037	0.0276 ± 0.0028	0.3699 ± 0.0068	0.0396 ± 0.0048
SMP (Baseline)	0.0126 ± 0.0020	0.0680 ± 0.0047	0.3330 ± 0.0059	0.1315 ± 0.0031

The narrative in Friesland is slightly different. Both frameworks degrade on SD and DARD relative to Utrecht and the full-province evaluation, indicating that both struggle to reproduce the distance and activity-purpose distributions of a rural province that was underrepresented in the training data. SI, however, remains stable, marginally improving across both approaches. This suggests that temporal trip spacing is structurally robust and generalises across the urban–rural division.

DailyLoc, however, degrades more drastically, falling from 0.0396 (± 0.0048) to 0.1329 (± 0.0119) for Llama, which brings SynTrav's previously superior score into line with the SMP. Notably, the SMP's DailyLoc mean remains flat across all evaluation contexts (0.1315, 0.1315 ,0.1309), meaning the baseline has settled on a stable but mediocre aggregate estimate of daily trip counts that holds across urbanisation levels without ever excelling.

Table 5.8 – LLM vs SMP Baseline for Friesland

Framework	SD	SI	DARD	DailyLoc
Llama 3.3-70B	0.0449 ± 0.0061	0.0220 ± 0.0036	0.4413 ± 0.0070	0.1329 ± 0.0119
SMP (Baseline)	0.0259 ± 0.0025	0.0583 ± 0.0042	0.4064 ± 0.0103	0.1309 ± 0.0031

To assess whether the lower scores in Friesland are attributable to urbanisation level rather than architectural limitations, two maps were produced. These can be observe at Figure 5.13. The first shows the overall composite score achieved by the LLM across all provinces. It confirms that the LLM struggles in more rural areas, with Friesland scoring among the lowest in the country, in contrast with Utrecht, South Holland, and North Holland, which rank among the highest. The second map confirms the high urbanisation level of these three provinces.

The personas constructed from ODiN were predominantly shaped by urban and semi-urban travel behaviour. When applied to a province where trip distances are longer, activity options are fewer, and daily travel rhythms differ structurally, both frameworks show their limitations.

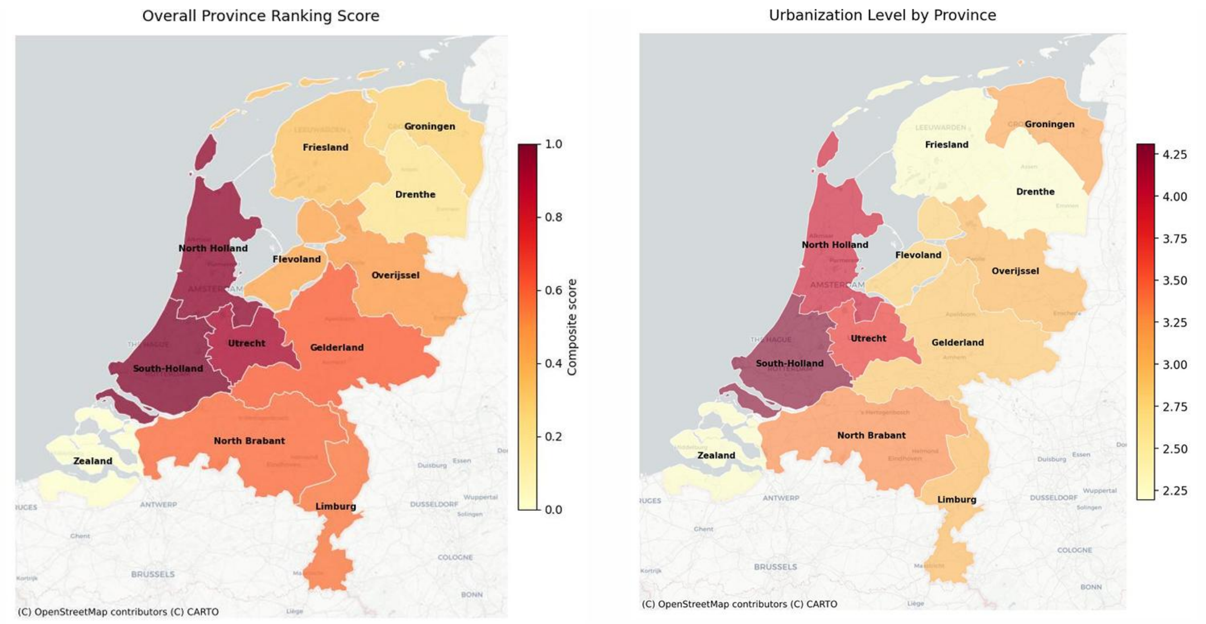


Figure 5.13– Province Map by Ranking Score and Urbanization Level

Taken together, these findings suggest that SynTrav's degradation in Friesland reflects a distributional limitation rather than an architectural one. SynTrav holds advantages in urban contexts where personas were more abundantly represented in the training data and patterns were more clearly defined. However, struggles in rural contexts where travellers exhibit distinctly different rhythms. Importantly, nothing in the personas or prompts was changed between the urban and the former benefits from richer and more representative training data. An interesting direction for future work would be enriching the persona construction with rural-representative respondents, or to supplement it via prompt injection, and to test whether this diminishes the gap under matched-sample conditions.

5.6 External Transferability

As discussed in Section 4.4, the synthetic population for Oeiras was conditioned on IMOB 2017 modal targets and Oeiras-specific commuting statistics when possible. Figure 5.7 allows a direct assessment of whether the LLM was able to reason about and shift population behaviour, despite having been trained on Dutch personas.

The largest shifts occur in car and cycling mode choice, consistent with the prompt injection. In addition to the IMOB 2017 targets, the LLM also received contextual information stating that Oeiras has higher car ownership relative to the Portuguese national average, and that usable cycling infrastructure is effectively absent.

Car use moved from 33.4% (NL) to 50.6% (PT), and cycling collapsed from 24.7% to 4.3% against a target of 0.5%. Both results show that the prompt injection redirected the model's modal prior in the correct direction, although the cycling share remains influenced by the strong cycling preference embedded in the Dutch baseline. Public transport moved in the right direction but undershot the target. Walking, conversely, barely needed to shift but ended up overestimated, likely because some of the suppressed cycling trips were reassigned to walking instead.

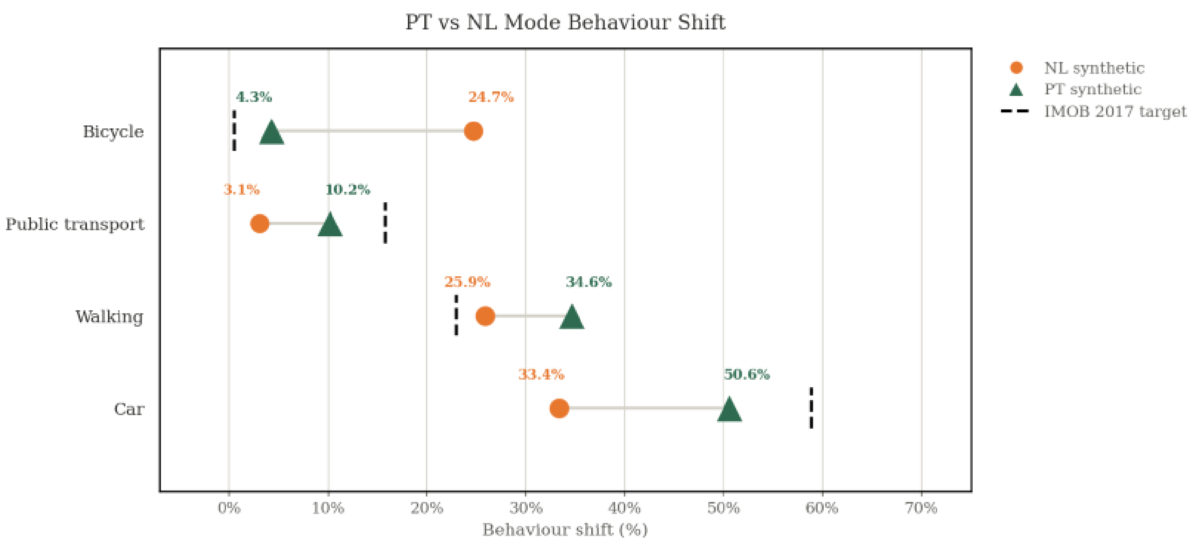


Figure 5.14– PT vs NL Mode Behaviour Shift

Looking now at the mode shift by purpose (Figure 5.8), it’s possible to assess whether the changes were intentional and coherent.

The changes for work trips were the most prominent. Car share for work trips jumped from 42% to 80.6%, while bicycle share dropped sharply from 27.6% to 0.2%, showing that the LLM aligned well with Oeiras's commuting structure (51% inter-municipal work trips, car-dominant via the A5 corridor).

For shopping trips, both car and walking increased significantly, at the expense of cycling, which fell from 32% to 0.2%. For leisure activities, walking became the dominant mode, rising from 24.4% to 50.7%, reflecting the injected context about coastal leisure destinations being reachable on foot. This is a case where the location-specific narrative (Estoril, coastal walkability) appears to have had more influence than the aggregate modal percentages alone and likely contribute to the overall overestimation of walking noted earlier.

Finally, education trips saw the largest increase in public transportation usage, from 11.2% to 35.7%, alongside a slight increase in car usage. Notably, cycling retains a higher share for education trips than for any other purpose, which is plausible because no Oeiras-specific context for education trips was provided, making it easier for the LLM to fall back on Dutch priors.

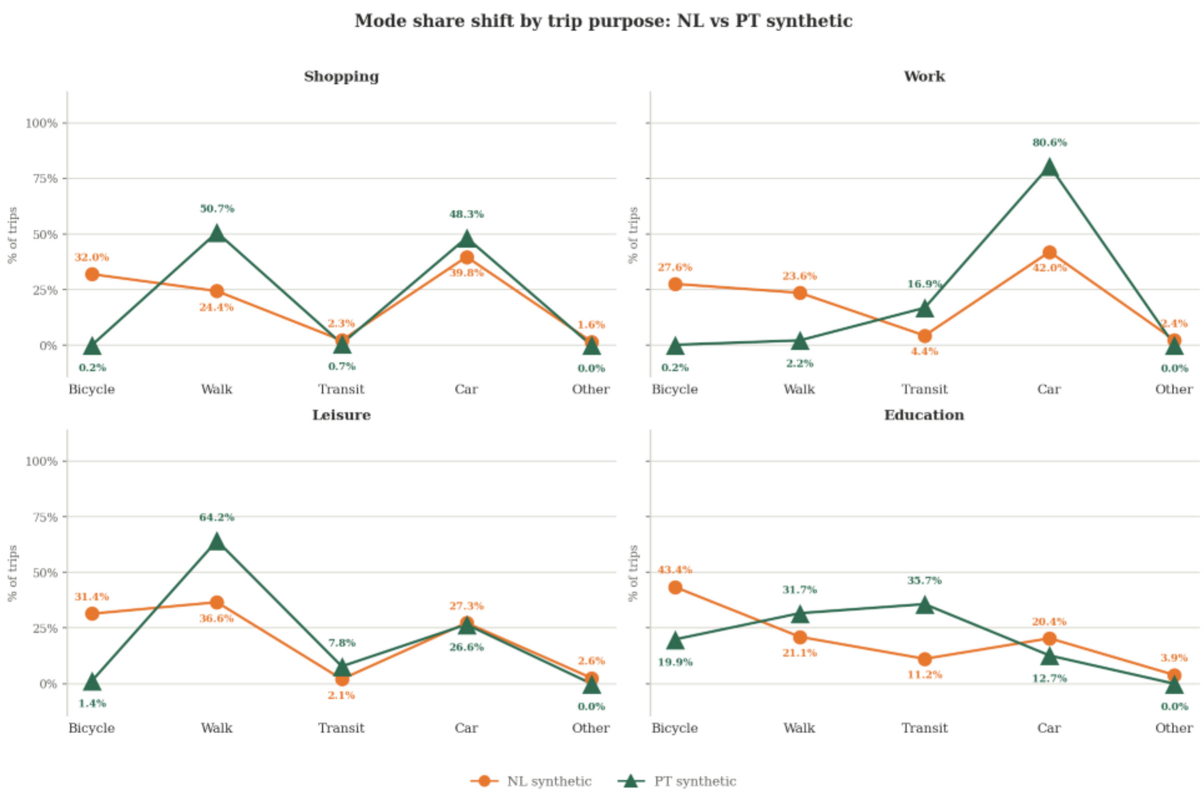


Figure 5.15– PT vs NL Mode Shift by Purpose

To understand how the distributions were actually affected, we can observe Figure 5.9.

On trip distance, the Portuguese synthetic population presents a more concentrated peak of short trips than the Dutch population, however, it also contains a secondary peak around 10–15 km. This generally aligns with the bimodal structure of Oeiras, (short intra-municipal errands of around 2.9 km versus longer inter-municipal commutes of 14–16 km). This concentration of short trips may also contribute to the walking overshoot noted earlier, since the injected context positioned Linha de Cascais as the realistic alternative for longer trips while scoping walking explicitly to short local journeys.

Regarding trip frequency, the Portuguese synthetic population concentrates more heavily around 1–3 trips per day, compared to the Dutch synthetic population's typical range of 2–4. This is consistent with the injected target of average 2.60 trips/day for the AML region.

The Step Interval (SI) metric shows a more right-shifted distribution for the Portuguese population, with longer gaps between trips than in the Dutch case, consistent with the inter-municipal trip and car-dominance context. In the temporal analysis, departure times for both populations coincide on the morning peak, but the afternoon return peak is more spread out for the Portuguese synthetic population. Since trip timing was not directly addressed in the prompt, these changes were not explicitly prompted and instead appear to be a consequence of the modal shift itself and prior model knowledge, where longer, car-dominant routes tend to introduce more variable journey durations than shorter cycling trips.

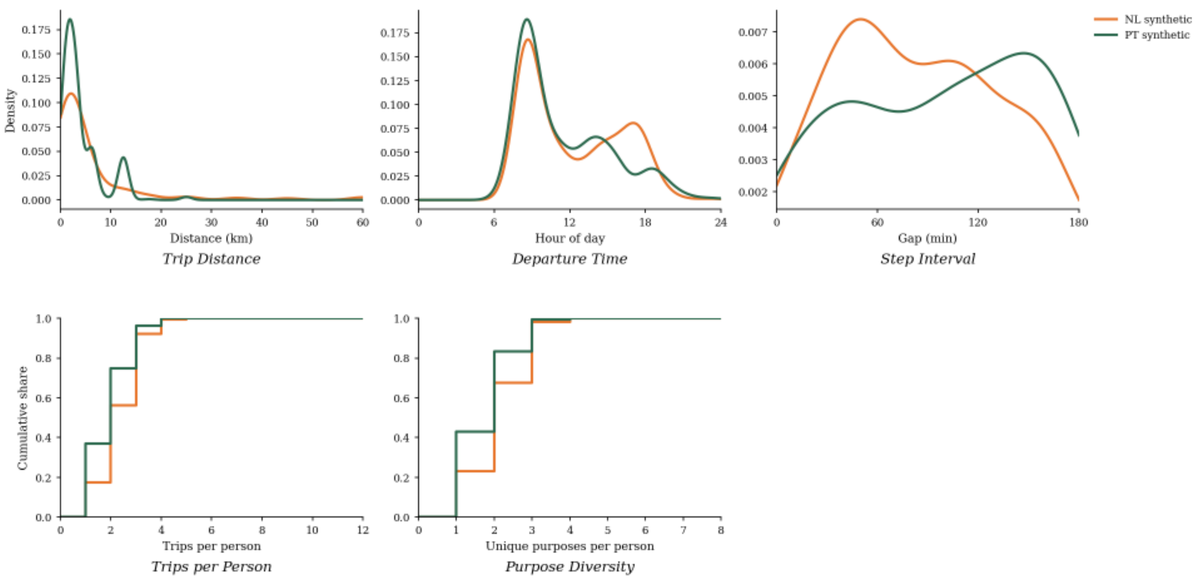


Figure 5.16– PT vs NL Distributions Comparison

As mentioned in Section 4.4, full spatial evaluation is not possible due to the lack of available comparable data. However, we assess qualitatively how well the shifted Portuguese behaviours translate onto a real map and if they remain geographically coherent.

Looking at both maps in the Figure 5.17, it’s possible to observe that employed agents cluster around an office hub (the Carnaxide–Linda-a-Velha business corridor near Taguspark) while unemployed agents shift toward parks, markets, and sports.

Employed travellers clustering around this corridor rather than along the coast is consistent with Taguspark having been injected as a non-walkable employment hub.

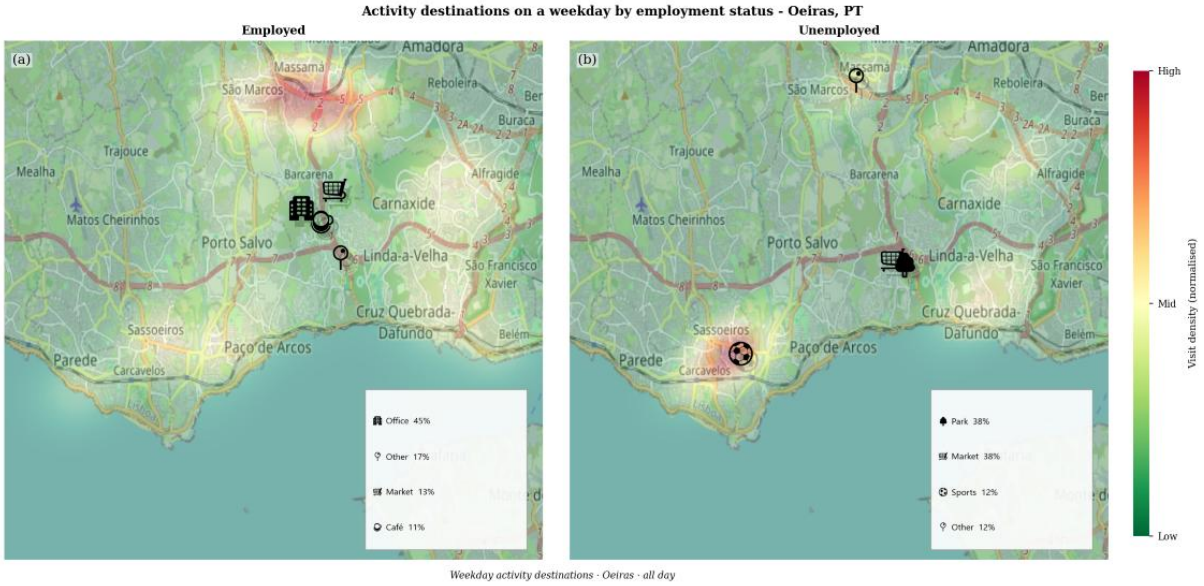


Figure 5.17 - Employed vs Unemployed Activities Destinations in Oeiras

Assessing whether the LLM-based generation framework was capable of redirecting its behaviour using prompt-injected Portuguese statistics, we observe that it could adapt meaningfully, even when trained exclusively on Dutch behavioural data.

There are instances where the adaptation overshot or undershot the intended targets, particularly for walking. However, directional correctness is observed across most modes and purposes, particularly in the car and cycling shifts driven by Oeiras's commuting structure, even though a residual cycling preference persists, stemming from distributional bias in the training corpus. For real-world deployment, both of these shortcomings could likely be addressed at the prompt level, without requiring model retraining.

5.6.1 PROMPT INJECTION SENSITIVE ANALYSIS

The results of the sensitive analysis can be observed in Figure 5.17

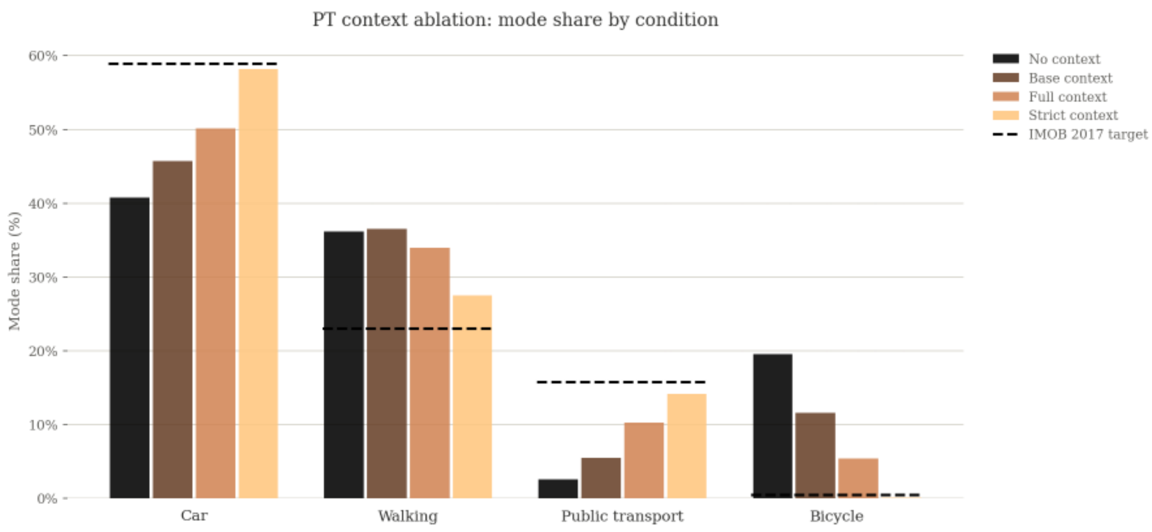


Figure 5.18– Portugal Prompt Ablations

Observing the four conditions, each additional layer of context moves the generated mode shares closer to the IMOB 2017 targets, with total absolute deviation falling from 63.6 pp (No Context) to 48.1 pp (Base), 30.2 pp (Full), and 7.2 pp (Strict). This total absolute deviation is defined here as the sum of absolute percentage-point differences between each condition's generated mode shares and the Itargets across the four modes.

The No Context variant, as expected, reflects most intensely the Dutch prior embedded in the persona patterns, though cycling usage decreases slightly relative to the Dutch baseline. Base Context reduces cycling substantially to 11.6%, and the soft cycling note in Full Context suppresses it further to 5.4%, while car use rises from 40.8% to 50.1% and public transport from 2.6% to 10.3%. Strict Context eliminates cycling almost entirely (0.1%) and yields the closest overall fit to IMOB 2017 targets (7.2 pp).

Despite this, Strict Context was not adopted as the primary condition. As shown in Chapter 5.6, the explicit no-cycling directive pushes both distance and temporal distributions back toward the Dutch baseline, erasing the behavioural divergence the transfer is meant to demonstrate. It was also prioritised to preserve the model's freedom to reason from the injected statistics without being explicitly instructed. Furthermore, as shown in the previous analysis, walking is the only mode that diverges from its target under Full Context, and this divergence intensifies under.

6. PRACTICAL IMPLICATIONS

It is widely documented in the literature that travel surveys are expensive, often difficult to access, frequently proprietary, and time-intensive to conduct. Synthetic data, in its various forms, has emerged as a possible solution. (Amin et al., 2025; Z. Li et al., 2023; Nadas et al., 2025)

Traditional methods, discussed earlier in the literature review, have attempted to address this issue, but typically require retraining when applied to unseen contexts, leaving them highly dependent on the same costly survey data (Qin et al., 2026).

LLMs offer an alternative path. Beyond sometimes outperforming classical frameworks in capturing behaviour better (X. Li et al., 2024; J. Wang et al., 2024), they also reduce reliance on continuous travel surveys. Rather than adjusting model weights through retraining, behavioural context can be supplied directly through the prompt grounding an entire diary generation framework in publicly available census and land-use data rather than proprietary travel survey data, while still achieving plausible results, in some cases exceeding the aggregate realism of classical benchmarks trained directly on the target data. (*Amin et al., 2025*)

As also discussed in the literature review, X. Li et al., (2024) successfully demonstrated the adaptation of travel diaries to a COVID-19 prompt injection without any retraining. In this thesis, we showed that an LLM-based framework trained on Dutch data could be redirected toward Portuguese behaviour using prompt-injected statistics alone, with shifts occurring in the correct direction. This responsiveness matters for a second reason beyond cost. Travel behaviour is not static, it shifts with economic conditions, infrastructure changes and policy interventions (*Amin* et al 2025). A model whose behaviour can only be updated through retraining on new survey data is structurally slow to respond to such changes.

None of this positions LLM-based generation as a replacement for travel surveys where they exist. Rather, it is a complementary tool. Where survey data is available, it remains the strongest source of calibration. Where it is unavailable or outdated, prompt-based context injection offers a way to approximate updated behaviour at a fraction of the cost of new data collection.

The true cost of a national household travel survey is difficult to access reliably. However, drawing on an earlier study that examined this issue, this section provides an estimate scaled to ODiN's sample size, for comparison with the cost of LLM-based generation. These figures should not be read as the actual cost of ODiN, which is not publicly disclosed. They are merely serving as an indication of the scale of investment that a survey of this size typically requires

(Stopher & Greaves, 2007) report two relevant values from the Sydney Household Travel Survey. Face-to-face interviews cost approximately AUD 350 per household, while partially automated GPS-based methods cost around USD 170. These are 2006 figures. The authors note that declining response rates and rising labour costs mean current costs are likely higher, although they also note that online surveys, as ODiN is, tend to be cheaper. A more recent reference, the 2024 US NextGen NHTS, point that reports participant incentives alone are $20–22 per household.

Scaling the Stopher and Greaves figures to ODiN's size (61,953 respondents, approximately 29,500 households at an average Dutch household size of 2.1) gives an estimated fieldwork cost on the order of €9–16 million in 2026 terms.

By contrast, generating the full 565-agent population in this thesis using Llama 3.3-70B via the Groq API cost approximately $2.86–2.90. Because the model is open-source, running it locally would reduce the costs even more. Due to local hardware limitations, generation in this case was outsourced to an external machine.

Scaling to ODiN's size:

$$\sim 62\,000\ respondents \ \div 570 = 109\ times\ \text{experimental population}$$
$$2.90 * 110 \ \approx 316$$

Equation (6.1)

This would cost roughly a few hundred dollars via API. These values are also estimates, since the cost may not scale exactly proportionally with the number of agents, however, the gap between the two approaches remains substantial.

7. CONCLUSIONS AND FUTURE RESEARCH

7.1 Conclusions

This thesis proposed SynTrav, a travel diary generation framework that adapts longitudinal LLM-agent approaches to the single-day structure of ODiN, the Dutch national travel survey from 2022. Focused on facilitating the LLM's ability to capture the fundamentals and dependencies behind mobility and nuanced individual attributes, personas were constructed from socio-demographic groupings, which condition both the pattern extraction stage and a recursive daily-plan generation pipeline. A central aim of this study was to test whether this framework could generalise outside it's original context, including a cross-national transfer to Portugal, where prompt-injected statistics were used to redirect the framework's behavioural priors without retraining.

The first research question (**RQ 1**) asked whether an LLM agent framework, conditioned on single-day self-reported survey data, could produce realistic synthetic travel diaries. The answer is yes. The framework generated diaries that are sensitive to persona socio-demographics and behaviourally plausible across multiple dimensions, while remaining spatially allocatable to a real urban environment, though not without limitations.

One of the central challenges in building this framework stemmed from the nature of the data itself. In the absence of individual travel histories, statistical, motivational, and daily-plan prompts were introduced to provide the LLM with sufficient contextual background. The ablation studies confirm the preference for the Full framework, not because it leads on every isolated metric, but because it is the only configuration that keeps all four evaluation metrics jointly within realistic ranges. The same reasoning applies to mode and distance. Testing showed that the LLM cannot faithfully reproduce either attribute through reasoning alone, so both are anchored to observed survey data to preserve a fidelity the model does not achieve on its own. The final configuration retains all components and reasons freely about purpose, motivation, and coherence.

When compared to the Semi-Markov Process baseline, SynTrav demonstrates superior temporal coherence and more accurate daily trip budgets, by large margins than those by which the baseline leads on trip distance and activity-purpose distributions. Spatial allocation, evaluated separately via STVD, favours the baseline (0.28 versus 0.38), reflecting its direct statistical fit to observed destination patterns.

Sixty-two respondents were asked to judge 16 travel diaries, rating the plausibility and classifying it as real or synthetic. The evaluators struggled to identify synthetic diaries as generated, rating them at least as plausible as real ones, with persona–purpose consistency emerging as the most reliable cue for synthetic detection. The overall accuracy was 46% (460/992; 95% CI [43.2%, 49.5%]), below the 50% chance level.

Addressing **RQ 2** (internal transferability), SynTrav generalises better to unseen provinces with high urbanisation levels, but performance degrades for more rural contexts. Both frameworks struggled with the rural province, though SynTrav showed greater sensitivity to this limitation.

Finally, addressing whether the LLM was able to reason about and shift population behaviour despite being trained on Dutch personas (**RQ 3**). Prompt-injected was sufficient to redirect the behaviour across the major of modes and purposes, particularly for car and cycling use, although a residual cycling preference attributable to the training corpus persisted. Some targets, specially walking, were over- or undershot. These behavioural shifts varied coherently according to persona profile rather than being applied uniformly across the population, indicating that the framework preserved its capacity to associate individual characteristics with mobility behaviour under the new cultural context. Spatial allocation for Oeiras, though exploratory, demonstrated that the framework can be geographically grounded in a new country without retraining the spatial allocation logic itself.

In sum, the results demonstrate that the framework was able to produce plausible travel diaries sensitive to persona profiles, though not flawlessly. The main tensions identified relate primarily to the representativeness of the behavioural reference data and the completeness of the injected context. The behavioural-shift results showcase promising capabilities for LLM agents to support urban mobility analysis and reduce data-collection costs, positioning the framework as a complementary tool alongside traditional survey data.

7.2 Limitations and Future Research

As with the majority of studies, the design of the current framework is subject to limitations that highlight opportunities for further investigation.

An interesting extension of this work would be testing the framework against a longitudinal dataset, where travel histories for the same individual are available. The current framework compensates for the absence of personal history through group-level conditioning. With access to longitudinal data, a "past trajectories" block could instead be injected into the generation pipeline, allowing the framework to reason about habitual destinations, intra-personal variability, and individual routine stability. Comparing the two conditioning regimes under the same evaluation metrics would also quantify the accuracy cost of relying on group-level compensation alone.

Introducing household dynamic, as proposed by (Y. Liu et al., 2025), can also be a promising direction. This remains a relatively neglected area, despite the fact that household-level relationships meaningfully affect individual commuting choices. Additionally, introducing multi-modal routing would help bring the framework closer to a fully detailed real-world scenario.

A further limitation, already mentioned, is that the Portuguese transfer scenario could not be evaluated using the same distributional metrics applied to the Dutch case, due to the absence of comparable trip-level data. Future work could explore, through a more in-depth evaluation, whether full transferability via prompt injection is achievable.

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APPENDIX A

Table A.1- LLM Scores Variable Ranking

Variable	Score (mean score across groups)
Car Ownership	7.90
Walk frequency	7.80
PT frequency	7.70
Car driver frequency	7.60
Bike frequency	7.30
Urbanization	7.00
E-bike in household	6.90
Income	6.10
Driver license	5.95
Education	5.70
Family structure	5.50
Train frequency	5.30
Gender	4.60
Minor in household	4.60
Commute mode habit	3.10

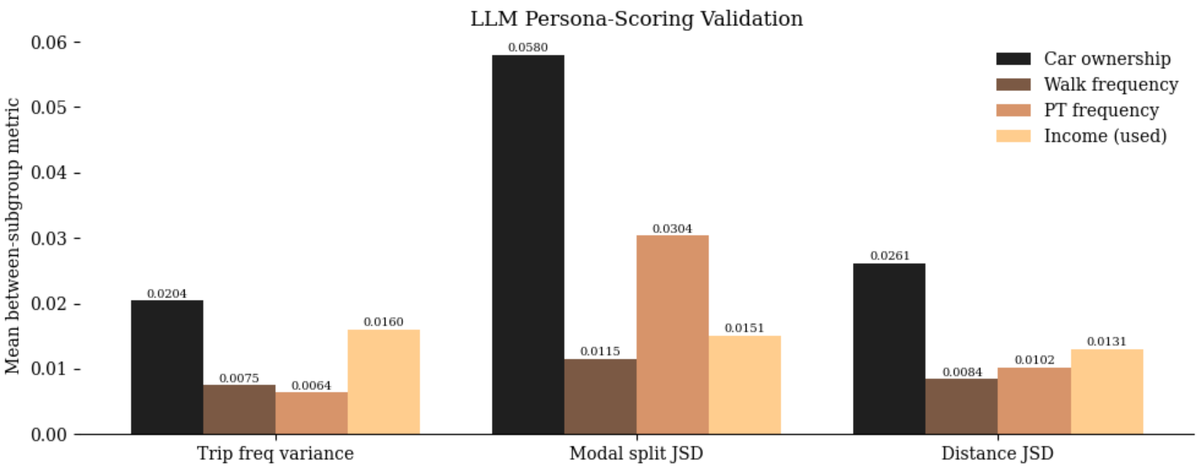

Figure A.1 - LLM Persona-Scoring Validation

Table A.2– Motivational Summary Output

Retired | 65+ | weekday | Income=Low income
Without employment obligations, this person's travel is driven by discretionary activities — social visits, errands, and leisure — distributed across the middle of the day when they feel most active. Their travel is constrained by limited car access and a preference for short distances, making walking and cycling the dominant modes for nearby destinations. The typical travel day is simple, often a single outbound trip and return, rather than a chained sequence of activities. Unlike the average Dutch traveller, this group concentrates nearly all movement within a narrow time window around midday and is significantly less likely to travel at all on any given day.

Prompt A1

Patterns Extraction Prompt

PATTERN_EXTRACTION_PROMPT = """

You are analysing Dutch daily mobility data from the ODiN 2022 national travel survey.

<GROUP PROFILE>

Group: {group_name}

N persons: {n_persons} | N trips: {n_trips} | Avg trips/person: {trips_per_person}

Demographics:

- Age: {age_dist}

- Gender: {gender_dist}

- Income: {income_dist}

- Household composition: {household_dist}

- Urbanisation: {urban_dist}

Travel behaviour distributions:

- Transport mode share: {mode_share}

- Trip purpose share: {purpose_share}

- Departure time share: {dep_time_share}

- Distance class share: {distance_share}

</GROUP PROFILE>

Using step-by-step reasoning:

1. For each demographic attribute (age, occupation, income, household), explain how it ikely shapes this group's travel choices.

2. Identify the dominant daily mobility rhythm: when they travel, how far, by what mode, and for what purpose.

3. Note any notable secondary patterns (e.g. a minority that deviates from the dominant mode).

4. Write a concise mobility pattern summary (3-5 sentences) describing what makes this group distinct from the broader Dutch travelling population.

Think through each step internally, then output ONLY the final 3-5 sentence mobility pattern summary.

"""

COT_VALIDATE_REFINE_PROMPT="""

You are analysing Dutch daily mobility data from the ODiN 2022 national travel survey

<IDENTIFIED PATTERN>

Group: {group_name}

{pattern}

</IDENTIFIED PATTERN>

<REAL TRAJECTORIES>

Below are anonymised trip sequences from ODiN respondents in this group.

Format: Departure time class | Transport mode | Trip purpose | Distance class

{trajectories}

</REAL TRAJECTORIES>

<MASKED TRAJECTORIES>

Below are trip sequences with one field per trip hidden as [MASKED].

Format: Departure time class | Transport mode | Trip purpose | Distance class

{masked_trajectories}

</MASKED TRAJECTORIES>

“””

Prompt A2

Motivational Summary Prompt

MOTIVATIONAL_SUMMARY_PROMPT = """

You are a daily planning expert, skilled at creating clear and reasonable daily plans for individuals based on various factors (such as location, external conditions, personal attributes, and previous reflections).

<INDIVIDUAL PROFILE>

{persona_profile}

</INDIVIDUAL PROFILE>

<MOBILITY PATTERNS>

{mobility_pattern}

</MOBILITY PATTERNS>

Write a motivational summary (4–5 sentences) that explains this person's travel behaviour on a typical {day_type}. Address in order:

1. The primary driver(s) of their travel — what obligations or activities bring them out of the home on this type of day.

2. The constraints that shape when and how they travel — time pressure, mode availability, household role, income, or physical capacity.

3. The dominant trip purposes for this person today and whether their travel day is typically simple (one main activity) or complex (a chain of activities).

4. Any behavioural tendency that distinguishes this group from an average Dutch traveller.

Write in third person. Be specific to this profile. Interpret the patterns — do not restate the statistics.

"""

Prompt A3

Daily Plan Prompt

DAILY_PLAN_PROMPT = """

You are a daily planning expert, skilled at creating realistic daily plans for individuals based on their personal profile and travel behaviour.

<MOTIVATIONAL SUMMARY>

{motivational_summary}

</MOTIVATIONAL SUMMARY>

<INDIVIDUAL PROFILE>

{persona_profile}

</INDIVIDUAL PROFILE>

<MOBILITY PATTERNS>

{mobility_pattern}

</MOBILITY PATTERNS>

Generate a plausible step-by-step daily plan for a typical {day_type}.

Trip count constraint: This person makes approximately {individual_trip_budget} trips today. Include no more than {individual_trip_budget} travel steps (outbound + return combined) in the plan. The motivational summary explains why and what kind of trips — use it to decide which trips to include, not to add more.

Constraints:

- Cover the full day from waking up to going to sleep; anchor the morning routine to the group's most common first departure time shown in the mobility patterns

- Trip chains are allowed (e.g. work → school → supermarket → home). Do not add a return-home step after every activity unless the person would genuinely go home in between.

- Every trip must use a mode drawn from the group's known modes in the mobility patterns; travel step descriptions must only reference those modes — do not invent modes not present in the mobility patterns

- The plan should not include any speculative or uncertain terms. If there are any unspecified contexts, you may make appropriate assumptions.

- Provide the plan directly, without explanations or summaries

Format each step strictly as:

[HH:MM] Activity

For any step involving physical movement, the activity MUST begin with one of these verbs:

Travel to / Walk to / Cycle to / Drive to / Take the train to / Return home by

Examples: [08:00] Travel to work | [17:30] Return home by car | [10:00] Walk to the supermarket

For all other steps (waking up, eating, working, resting), describe the activity directly.

"""

Prompt A4

Recursive Reasoning Prompt

RECURSIVE_REASONING_PROMPT = """Current time: {current_time}

Plan step: {plan_step}

<MOTIVATIONAL SUMMARY>

{motivational_summary}

</MOTIVATIONAL SUMMARY>

Daily plan overview:

{daily_plan}

Earlier schedule today:

{earlier_schedule}

<INDIVIDUAL PROFILE>

{persona_profile}

</INDIVIDUAL PROFILE>

<MOBILITY PATTERNS>

{mobility_pattern}

</MOBILITY PATTERNS>

The daily plan indicates this person is making a trip at {current_time}. Your task is to reason about the travel attributes for this specific trip using their profile, motivational summary, and group mobility patterns.

Plausibility check: If the earlier schedule shows this person has already returned home AND this plan step is a new outbound trip (not a return-home step), set PLAUSIBLE to NO and give a brief reason. In all other cases, set PLAUSIBLE to YES.

Available values — choose exactly one per field:

- Mode (pre-assigned based on group distribution): {assigned_mode}

- Purposes: {available_purposes}

- Destinations: {available_destinations}

Reply in exactly this format — no other text:

PLAUSIBLE: YES or NO

IMPLAUSIBILITY_REASON: [brief reason if PLAUSIBLE is NO, otherwise NONE]

MOTIVATION: [2 sentences — (1) why this person is making this specific trip given their profile and motivational summary, (2) how the assigned mode and group's mobility patterns support this choice]

PURPOSE: [exact value from Purposes list]

DESTINATION: [exact value from Destinations list]

"""

Prompt A5

Recursive Reasoning Prompt

PORTUGAL_CONTEXT_FULL = """

<LOCATION CONTEXT>

This person lives in Oeiras, Área Metropolitana de Lisboa (AML), Portugal.

Observed modal ground truth for AML (IMOB 2017 national travel survey):

- Car (driver + passenger): 58.9% of trips

- Walking (pedonal): ~23.0% of trips

- Public transport: 15.8% of trips

- Cycling: 0.5% of trips

- Average trips per day per mobile person: 2.60

- Dominant trip purposes: Work (30.8%), Shopping (19.8%)

AML mean one-way trip distances by purpose (IMOB 2017):

- Work: 29.5 min, 14.8 km

- Study: 23.6 min, 6.9 km

- Shopping: 16.7 min, 5.7 km

- Leisure: 28.0 min, 14.2 km

- Family accompaniment: 16.7 min, 7.1 km

- Personal matters: 16.9 km

- Intra-municipal trips (same municipality): mean 2.9 km in Oeiras

- Inter-municipal trips (crossing municipality): AML mean 16.4 km, 35 min

AML mean one-way trip distances by mode (IMOB 2017):

- Walking: 17 min, 1.5 km

- Bus (public): 45.8 min, 12.1 km

- Car (driver): 21.7 min, 12.8 km

- Car (passenger): 20.8 min, 13.4 km

- Train: 53.4 min, 19.1 km

- Metro: 39.7 min, 8.6 km

Oeiras commuting structure (Census 2021):

- 51.1% of employed Oeiras residents work in another municipality (mainly Lisbon), vs. 44.1% AML average and 34.1% national average

- Only 42.7% work within Oeiras itself; intra-municipal work trips are the minority

- Consequence: most work trips are cross-municipal and long (~14–16 km); intra-municipal trips average 2.9 km and are typically local errands or short leisure

- Oeiras is both a residential suburb and a major employment hub, drawing 94 inbound commuters per 100 residents (AML mean: 65)

Oeiras mobility context:

- Taguspark is the dominant employment hub in the municipality; it is not walkable from most residential areas and requires either a car or the Linha de Cascais suburban rail

- Linha de Cascais is the main public transport corridor, running along the coast through Oeiras to Lisbon

- The A5 motorway is the dominant car-commuting route to Lisbon

- Dedicated cycling infrastructure is minimal (fewer than 20 km of lanes across the municipality); cycling is not a viable mode for most trips

- Car ownership in Oeiras is high relative to the Portuguese national average

- Coastal leisure destinations (Estoril, Oeiras waterfront) are the primary recreation draw, reachable on foot from coastal residential areas or by Linha de Cascais

Location-specific constraints:

- usable cycling infrastructure is effectively absent

- For work, education, and service trips, prioritise car or public transport (Linha de Cascais) over walking

- Walking is appropriate only for short local trips (e.g. reaching a nearby shop, park, or rail station)

- For longer trips to Lisbon or along the coast, Linha de Cascais is a realistic alternative to car

</LOCATION CONTEXT>

"""

PORTUGAL_CONTEXT_SHORT = """

<LOCATION CONTEXT>

Location: Oeiras, AML, Portugal. Modal ground truth (IMOB 2017): Car 58.9% | Walking ~23.0% | Public transport 15.8% | Cycling 0.5%. Average 2.60 trips/day.

AML one-way trip distances by purpose (IMOB 2017): Work 14.8 km | Study 6.9 km | Shopping 5.7 km | Leisure 14.2 km | Family accompaniment 7.1 km | Personal 16.9 km

AML one-way trip distances by mode (IMOB 2017): Walking 1.5 km | Bus 12.1 km | Car 12.8 km | Train 19.1 km

Oeiras: 51% of workers commute to another municipality (mainly Lisbon); intra-municipal trips average 2.9 km.

usable cycling infrastructure is effectively absent. Walking is only appropriate for short local trips. For longer trips to Lisbon or along the coast, public transport (Linha de Cascais) is a realistic alternative to car.

</LOCATION CONTEXT>

"""

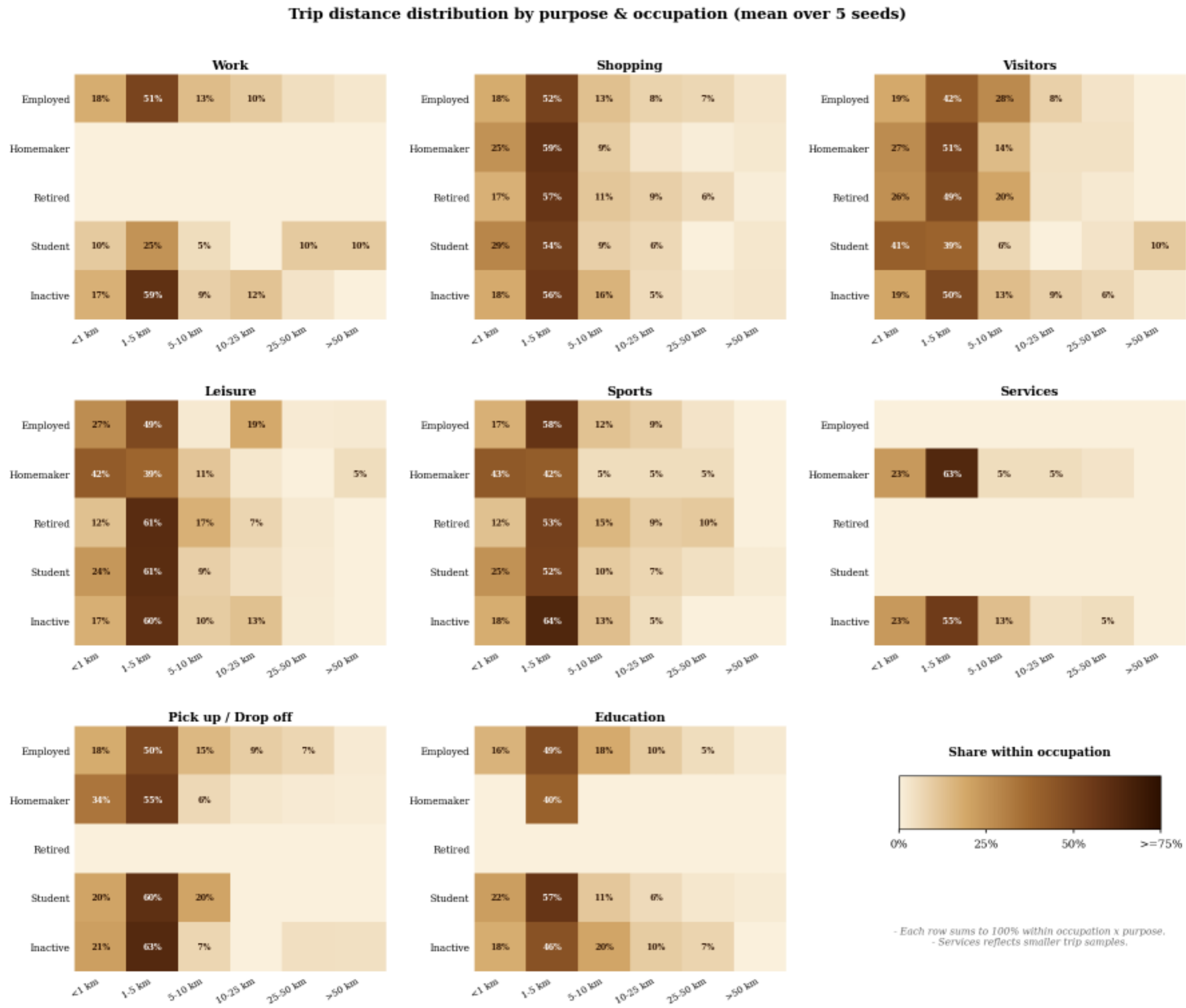

Figure A.2– Trip Distance by Purpose and Education

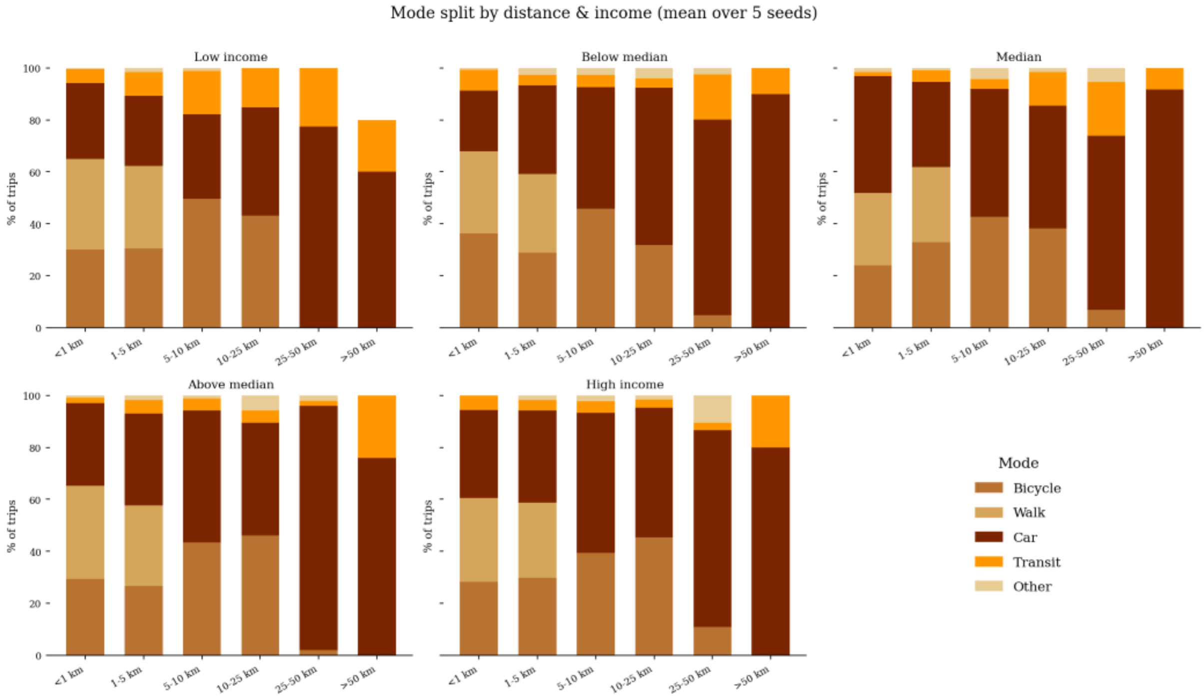

Figure A.3– Mode Split by Distance and Income

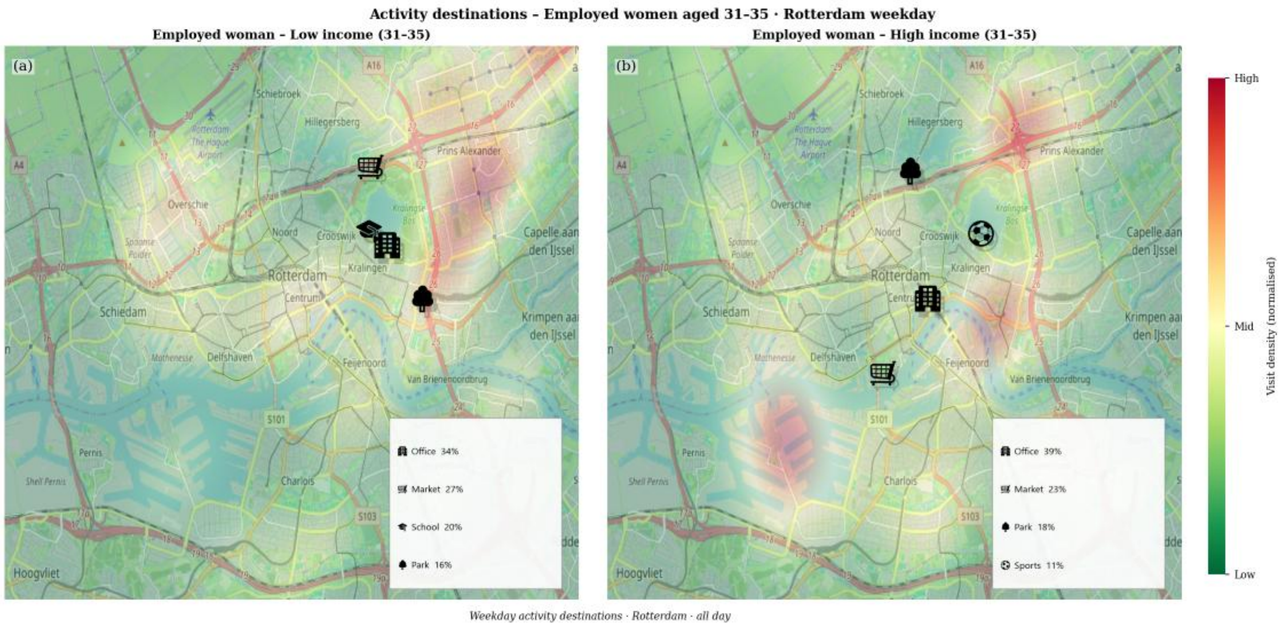

Figure A.4– Spatial Allocation (Low vs High Income) in Rotterdam

APPENDIX B

This Appendix serves to showcase one of the examples of a respondent from the human evaluation survey.

Diary 1

Inactive · age 26-30 · weekday

Time	Mode	Purpose	Distance
08:00	Walking	To and from work	1.8 km
16:00	E-bike	Sports/hobbies	3.1 km
18:00	E-bike	Sports/hobbies	3.1 km

Q1. How plausible is this as a real Dutch person's travel day? *

	1	2	3	4	5	
1 — Very implausible	○	○	◉	○	○	5 — Very plausible

Q2. If you rated below 3, what makes it implausible?

- [ ] Travel mode is unrealistic for this person
- [ ] Trip timing is implausible
- [ ] Trip sequence doesn't make sense
- [ ] Distances are unrealistic
- [ ] Too many or too few trips
- [ ] Purposes don't fit the profile
- [ ] Diary is too clean
- [ ] Nothing specific, just a gut feeling

Q3. Is this diary real or synthetic? *

- ◯ Real (ODiN survey)
- ◯ Synthetic (model-generated)

Q4. Optional comments on this diary

A sua resposta

Figure B.1 Survey Example

Data with Purpose.

Campus de Campolide, 1070-312 Lisboa, Portugal
novaims.unl.pt